

Cardiovascular

“As for me, except for an occasional heart attack, I feel as young as I ever did.”

—Robert Benchley

“Hearts will never be practical until they are made unbreakable.”

—The Wizard of Oz

“As the arteries grow hard, the heart grows soft.”

—H. L. Mencken

“Nobody has ever measured, not even poets, how much the heart can hold.”

—Zelda Fitzgerald

“Only from the heart can you touch the sky.”

—Rumi

“It is not the size of the man but the size of his heart that matters.”

—Evander Holyfield

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► CARDIOVASCULAR—EMBRYOLOGY

Heart embryology

EMBRYONIC STRUCTURE

GIVES RISE TO

Truncus arteriosus	Ascending aorta and pulmonary trunk
Bulbus cordis	Smooth parts (outflow tract) of left and right ventricles
Endocardial cushion	Atrial septum, membranous interventricular septum; AV and semilunar valves
Primitive atrium	Trabeculated part of left and right atria
Primitive ventricle	Trabeculated part of left and right ventricles
Primitive pulmonary vein	Smooth part of left atrium
Left horn of sinus venosus	Coronary sinus
Right horn of sinus venosus	Smooth part of right atrium (sinus venarum)
Right common cardinal vein and right anterior cardinal vein	Superior vena cava (SVC)

Heart morphogenesis

First functional organ in vertebrate embryos; beats spontaneously by week 4 of development.

Cardiac looping

Primary heart tube loops to establish left-right polarity; begins in week 4 of gestation.

Defect in left-right dynein (involved in L/R asymmetry) can lead to dextrocardia, as seen in Kartagener syndrome (primary ciliary dyskinesia).

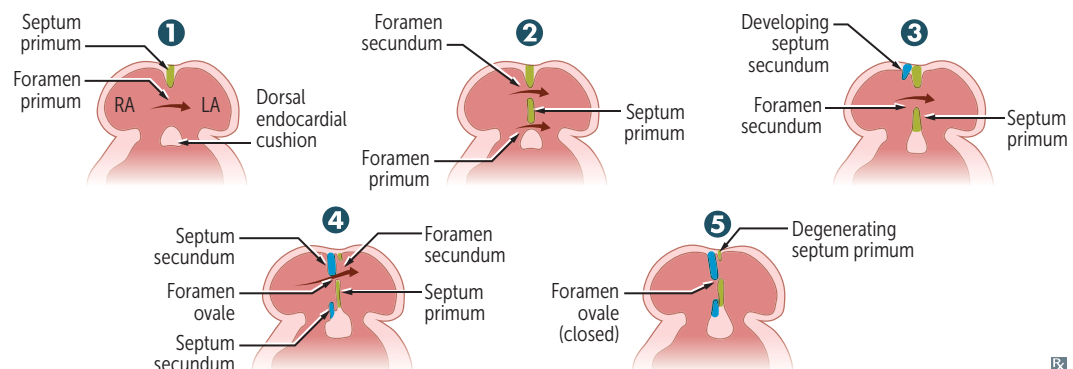
Septation of the chambers

Atria

- 1 Septum primum grows toward endocardial cushions, narrowing foramen primum.
- 2 Foramen secundum forms in septum primum (foramen primum disappears).
- 3 Septum secundum develops as foramen secundum maintains right-to-left shunt.
- 4 Septum secundum expands and covers most of the foramen secundum. The residual foramen is the foramen ovale.
- 5 Remaining portion of septum primum forms valve of foramen ovale.

6. (Not shown) Septum secundum and septum primum fuse to form the atrial septum.
7. (Not shown) Foramen ovale usually closes soon after birth because of ↑ LA pressure.

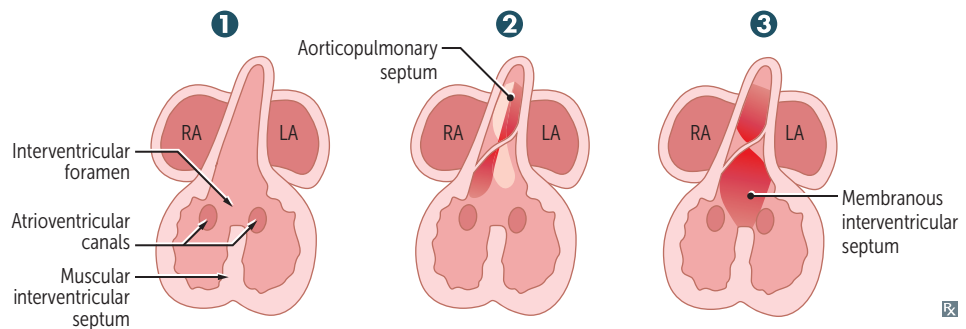
Patent foramen ovale—caused by failure of septum primum and septum secundum to fuse after birth; most are left untreated. Can lead to paradoxical emboli (venous thromboemboli that enter systemic arterial circulation), similar to those resulting from an ASD.



Heart morphogenesis (continued)**Ventricles**

- ❶ Muscular interventricular septum forms. Opening is called interventricular foramen.
- ❷ Aorticopulmonary septum rotates and fuses with muscular ventricular septum to form membranous interventricular septum, closing interventricular foramen.
- ❸ Growth of endocardial cushions separates atria from ventricles and contributes to both atrial septation and membranous portion of the interventricular septum.

Ventricular septal defect—most common congenital cardiac anomaly, usually occurs in membranous septum.

**Outflow tract formation**

Neural crest and endocardial cell migrations
 → truncal and bulbar ridges that spiral and fuse to form aorticopulmonary septum
 → ascending aorta and pulmonary trunk.

Conotruncal abnormalities associated with failure of neural crest cells to migrate:

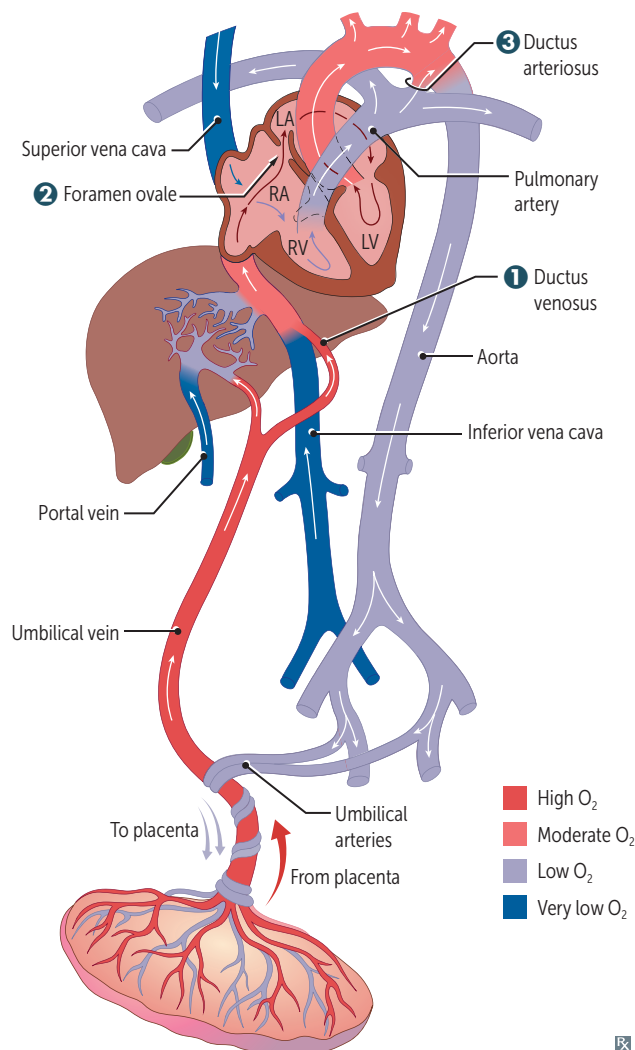
- Transposition of great vessels.
- Tetralogy of Fallot.
- Persistent truncus arteriosus.

Valve development

Aortic/pulmonary: derived from endocardial cushions of outflow tract.
 Mitral/tricuspid: derived from fused endocardial cushions of the AV canal.

Valvular anomalies may be stenotic, regurgitant, atretic (eg, tricuspid atresia), or displaced (eg, Ebstein anomaly).

Fetal circulation



Blood in umbilical vein has a PO_2 of ≈ 30 mm Hg and is $\approx 80\%$ saturated with O_2 . Umbilical arteries have low O_2 saturation.

3 important shunts:

- 1 Blood entering fetus through the umbilical vein is conducted via the **ductus venosus** into the IVC, bypassing hepatic circulation.
- 2 Most of the highly **O**xxygenated blood reaching the heart via the IVC is directed through the **foramen O**vale and pumped into the aorta to supply the head and body.
- 3 **D**eoxygenated blood from the SVC passes through the RA → RV → main pulmonary artery → **D**uctus arteriosus → **D**escending aorta; shunt is due to high fetal pulmonary artery resistance (due partly to low O_2 tension).

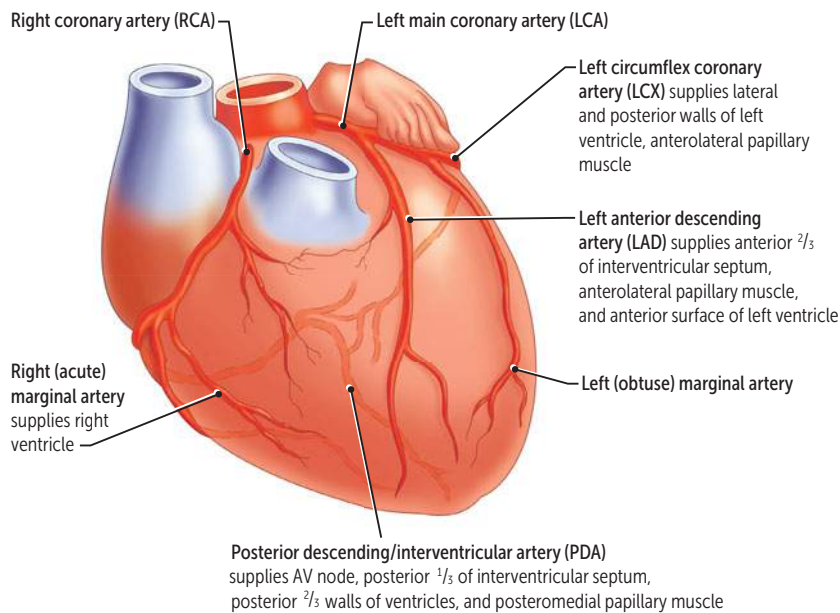
At birth, infant takes a breath; ↓ resistance in pulmonary vasculature → ↑ left atrial pressure vs right atrial pressure; foramen ovale closes (now called fossa ovalis); ↑ in O_2 (from respiration) and ↓ in prostaglandins (from placental separation) → closure of ductus arteriosus.

Indomethacin helps close PDA → ligamentum arteriosum (remnant of ductus arteriosus). Prostaglandins **E**₁ and **E**₂ **kEEp** PDA open.

Fetal-postnatal derivatives

Alla N tois → urachus	Media N umbilical ligament	Urachus is part of allantoic duct between bladder and umbilicus.
Ductus arteriosus	Ligamentum arteriosum	
Ductus venosus	Ligamentum venosum	
Foramen ovale	Fossa ovalis	
Notochord	Nucleus pulposus	
Umbi L ical arteries	Media L umbilical ligaments	
Umbilical vein	Ligamentum teres hepatis (round ligament)	Contained in falciform ligament.

► CARDIOVASCULAR—ANATOMY

Anatomy of the heart

SA and AV nodes are supplied by branches of the RCA. Infarct may cause nodal dysfunction (bradycardia or heart block).

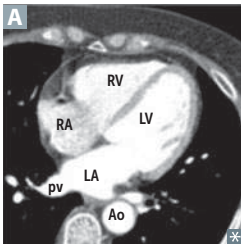
Right-dominant circulation (85%) = PDA arises from RCA.

Left-dominant circulation (8%) = PDA arises from LCX.

Codominant circulation (7%) = PDA arises from both LCX and RCA.

Coronary artery occlusion most commonly occurs in the LAD.

Coronary blood flow peaks in early diastole.



The most posterior part of the heart is the left atrium **A**; enlargement can cause dysphagia (due to compression of the esophagus) or hoarseness (due to compression of the left recurrent laryngeal nerve, a branch of the vagus).

Pericardium consists of 3 layers (from outer to inner):

- Fibrous pericardium
- Parietal layer of serous pericardium
- Visceral layer of serous pericardium

Pericardial cavity lies between parietal and visceral layers.

Pericardium innervated by phrenic nerve. Pericarditis can cause referred pain to the shoulder.

► CARDIOVASCULAR—PHYSIOLOGY

Cardiac output

$CO = \text{stroke volume (SV)} \times \text{heart rate (HR)}$

Fick principle:

$$CO = \frac{\text{rate of O}_2 \text{ consumption}}{\text{arterial O}_2 \text{ content} - \text{venous O}_2 \text{ content}}$$

Mean arterial pressure (MAP) = $CO \times \text{total peripheral resistance (TPR)}$

$MAP = \frac{2}{3} \text{ diastolic pressure} + \frac{1}{3} \text{ systolic pressure}$

Pulse pressure = $\text{systolic pressure} - \text{diastolic pressure}$

Pulse pressure is proportional to SV, inversely proportional to arterial compliance.

$SV = \text{end-diastolic volume (EDV)} - \text{end-systolic volume (ESV)}$

During the early stages of exercise, CO is maintained by $\uparrow$ HR and $\uparrow$ SV. During the late stages of exercise, CO is maintained by $\uparrow$ HR only (SV plateaus).

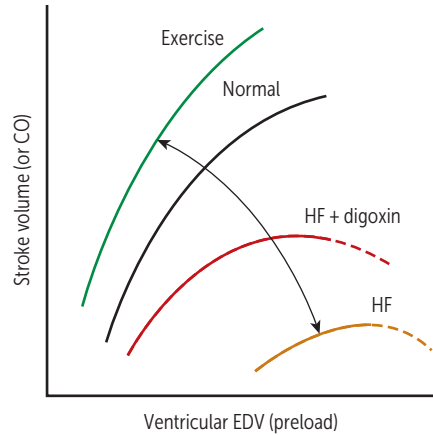
Diastole is preferentially shortened with $\uparrow$ HR; less filling time $\rightarrow$ $\downarrow$ CO (eg, ventricular tachycardia).

$\uparrow$ pulse pressure in hyperthyroidism, aortic regurgitation, aortic stiffening (isolated systolic hypertension in elderly), obstructive sleep apnea ($\uparrow$ sympathetic tone), exercise (transient).

$\downarrow$ pulse pressure in aortic stenosis, cardiogenic shock, cardiac tamponade, advanced heart failure (HF).

Cardiac output variables

Stroke volume	Stroke Volume affected by Contractility, Afterload, and Preload. ↑ SV with: ▪ ↑ Contractility (eg, anxiety, exercise) ▪ ↑ Preload (eg, early pregnancy) ▪ ↓ Afterload	SV CAP. A failing heart has ↓ SV (systolic and/or diastolic dysfunction)
Contractility	Contractility (and SV) ↑ with: ▪ Catecholamine stimulation via β_1 receptor: ▪ Ca^{2+} channels phosphorylated → ↑ Ca^{2+} entry → ↑ Ca^{2+}-induced Ca^{2+} release and ↑ Ca^{2+} storage in sarcoplasmic reticulum ▪ Phospholamban phosphorylation → active Ca^{2+} ATPase → ↑ Ca^{2+} storage in sarcoplasmic reticulum ▪ ↑ intracellular Ca^{2+} ▪ ↓ extracellular Na^+ (↓ activity of $\text{Na}^+/\text{Ca}^{2+}$ exchanger) ▪ Digitalis (blocks Na^+/K^+ pump → ↑ intracellular Na^+ → ↓ $\text{Na}^+/\text{Ca}^{2+}$ exchanger activity → ↑ intracellular Ca^{2+})	Contractility (and SV) ↓ with: ▪ β_1-blockade (↓ cAMP) ▪ HF with systolic dysfunction ▪ Acidosis ▪ Hypoxia/hypercapnia (↓ PO_2/↑ PCO_2) ▪ Non-dihydropyridine Ca^{2+} channel blockers
Myocardial oxygen demand	MyoCARDial O_2 demand is ↑ by: ▪ ↑ Contractility ▪ ↑ Afterload (proportional to arterial pressure) ▪ ↑ heart Rate ▪ ↑ Diameter of ventricle (↑ wall tension)	Wall tension follows Laplace's law: Wall tension = pressure × radius Wall stress = $\frac{\text{pressure} \times \text{radius}}{2 \times \text{wall thickness}}$
Preload	Preload approximated by ventricular EDV; depends on venous tone and circulating blood volume.	Venous vasodilators (eg, nitroglycerin) ↓ prEload.
Afterload	Afterload approximated by MAP. ↑ afterload → ↑ pressure → ↑ wall tension per Laplace's law. LV compensates for ↑ afterload by thickening (hypertrophy) in order to ↓ wall tension.	Arterial vasodilators (eg, hydrAzine) ↓ Afterload. ACE inhibitors and ARBs ↓ both preload and afterload. Chronic hypertension (↑ MAP) → LV hypertrophy.
Ejection fraction	$\text{EF} = \frac{\text{SV}}{\text{EDV}} = \frac{\text{EDV} - \text{ESV}}{\text{EDV}}$ Left ventricular EF is an index of ventricular contractility.	EF ↓ in systolic HF. EF normal in HF with preserved ejection fraction (HFpEF).

Starling curve

Force of contraction is proportional to end-diastolic length of cardiac muscle fiber (preload).

↑ contractility with catecholamines, positive inotropes (eg, digoxin).

↓ contractility with loss of myocardium (eg, MI), β-blockers (acutely), non-dihydropyridine Ca²⁺ channel blockers, dilated cardiomyopathy.

Resistance, pressure, flow

$$\Delta P = Q \times R$$

Similar to Ohm's law: $\Delta V = IR$

Volumetric flow rate (Q) = flow velocity (v) × cross-sectional area (A)

Resistance

$$= \frac{\text{driving pressure } (\Delta P)}{\text{flow } (Q)} = \frac{8\eta \text{ (viscosity)} \times \text{length}}{\pi r^4}$$

Total resistance of vessels in series:

$$R_T = R_1 + R_2 + R_3 \dots$$

Total resistance of vessels in parallel:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots$$

Capillaries have highest total cross-sectional area and lowest flow velocity.

Pressure gradient drives flow from high pressure to low pressure.

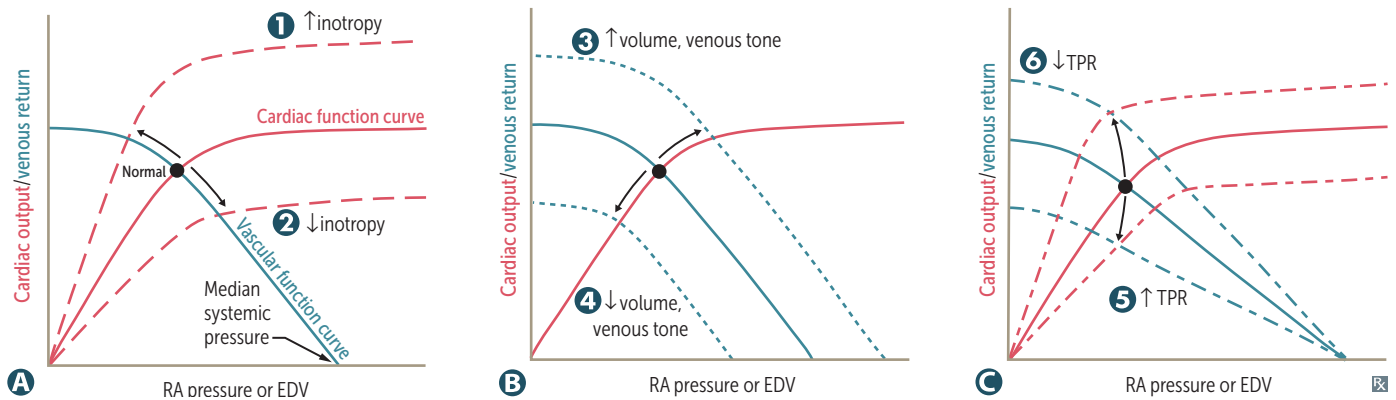
Arterioles account for most of TPR. Veins provide most of blood storage capacity.

Viscosity depends mostly on hematocrit.

Viscosity ↑ in hyperproteinemic states (eg, multiple myeloma), polycythemia.

Viscosity ↓ in anemia.

Cardiac and vascular function curves

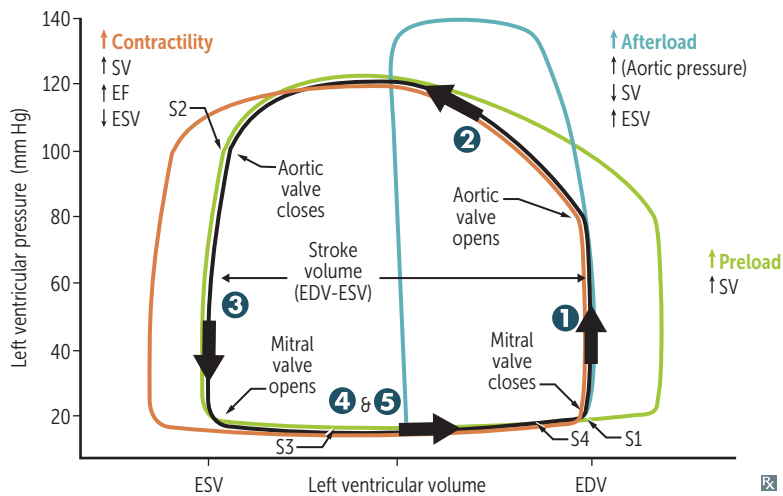


Intersection of curves = operating point of heart (ie, venous return and CO are equal).

GRAPH	EFFECT	EXAMPLES
A Inotropy	Changes in contractility → altered CO for a given RA pressure (preload).	1 Catecholamines, digoxin ⊕ 2 Uncompensated HF, narcotic overdose, sympathetic inhibition ⊖
B Venous return	Changes in circulating volume or venous tone → altered RA pressure for a given CO. Mean systemic pressure (x-intercept) changes with volume/venous tone.	3 Fluid infusion, sympathetic activity ⊕ 4 Acute hemorrhage, spinal anesthesia ⊖
C Total peripheral resistance	At a given mean systemic pressure (x-intercept) and RA pressure, changes in TPR → altered CO.	5 Vasopressors ⊕ 6 Exercise, AV shunt ⊖

Changes often occur in tandem, and may be reinforcing (eg, exercise ↑ inotropy and ↓ TPR to maximize CO) or compensatory (eg, HF ↓ inotropy → fluid retention to ↑ preload to maintain CO).

Pressure-volume loops and cardiac cycle



The black loop represents normal cardiac physiology.

Phases—left ventricle:

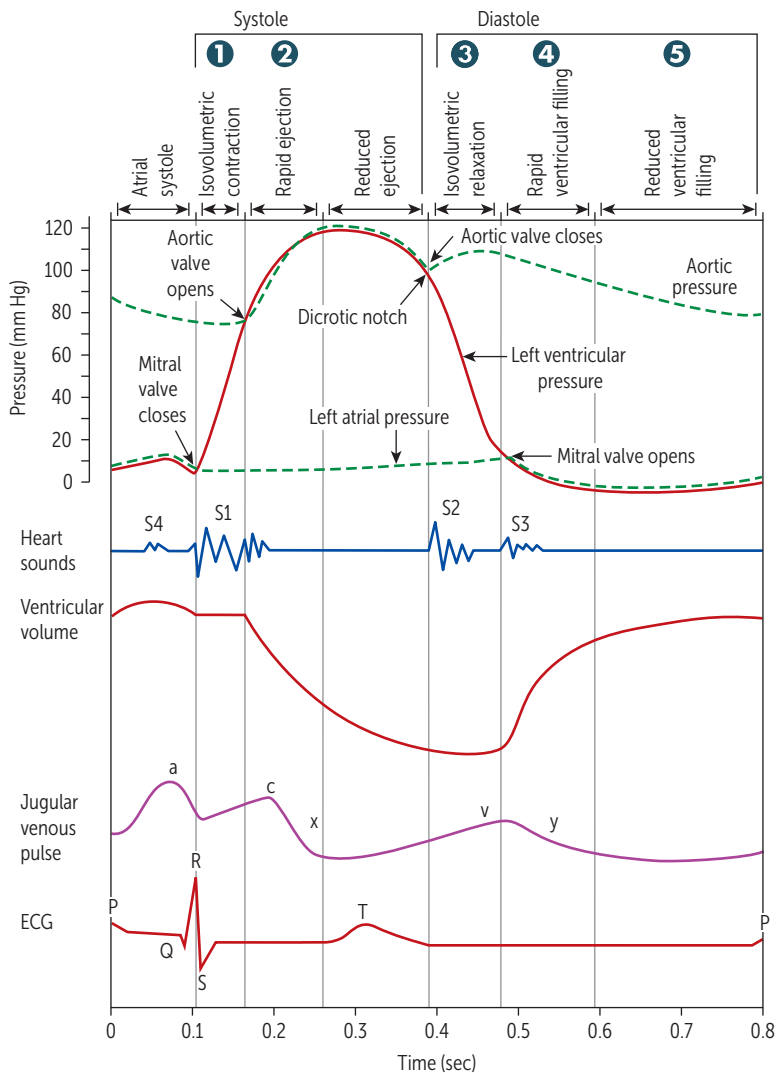
- 1 Isovolumetric contraction—period between mitral valve closing and aortic valve opening; period of highest O_2 consumption
- 2 Systolic ejection—period between aortic valve opening and closing
- 3 Isovolumetric relaxation—period between aortic valve closing and mitral valve opening
- 4 Rapid filling—period just after mitral valve opening
- 5 Reduced filling—period just before mitral valve closing

Heart sounds:

- S1—mitral and tricuspid valve closure. Loudest at mitral area.
- S2—aortic and pulmonary valve closure. Loudest at left upper sternal border.
- S3—in early diastole during rapid ventricular filling phase. Associated with ↑ filling pressures (eg, mitral regurgitation, HF) and more common in dilated ventricles (but can be normal in children and young adults).
- S4—in late diastole (“atrial kick”). Best heard at apex with patient in left lateral decubitus position. High atrial pressure. Associated with ventricular noncompliance (eg, hypertrophy). Left atrium must push against stiff LV wall. Consider abnormal, regardless of patient age.

Jugular venous pulse (JVP):

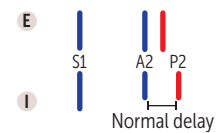
- a** wave—atrial contraction. Absent in atrial fibrillation (AF).
- c** wave—RV contraction (closed tricuspid valve bulging into atrium).
- x** descent—downward displacement of closed tricuspid valve during rapid ventricular ejection phase. Reduced or absent in tricuspid regurgitation and right HF because pressure gradients are reduced.
- v** wave—↑ right atrial pressure due to filling (“villing”) against closed tricuspid valve.
- y** descent—RA emptying into RV. Prominent in constrictive pericarditis, absent in cardiac tamponade.



Splitting

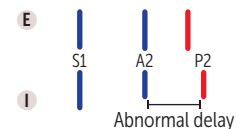
Normal splitting

Inspiration → drop in intrathoracic pressure
 → ↑ venous return → ↑ RV filling → ↑ RV stroke volume → ↑ RV ejection time
 → delayed closure of pulmonic valve.
 ↓ pulmonary impedance (↑ capacity of the pulmonary circulation) also occurs during inspiration, which contributes to delayed closure of pulmonic valve.



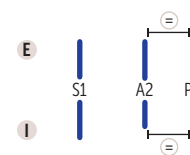
Wide splitting

Seen in conditions that delay RV emptying (eg, pulmonic stenosis, right bundle branch block). Causes delayed pulmonic sound (especially on inspiration). An exaggeration of normal splitting.



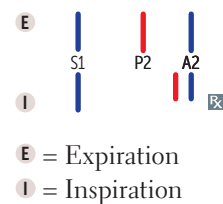
Fixed splitting

Heard in ASD. ASD → left-to-right shunt
 → ↑ RA and RV volumes → ↑ flow through pulmonic valve such that, regardless of breath, pulmonic closure is greatly delayed.



Paradoxical splitting

Heard in conditions that delay aortic valve closure (eg, aortic stenosis, left bundle branch block). Normal order of valve closure is reversed so that P2 sound occurs before delayed A2 sound. Therefore on inspiration, P2 closes later and moves closer to A2, thereby “paradoxically” eliminating the split (usually heard in expiration).



Auscultation of the heart

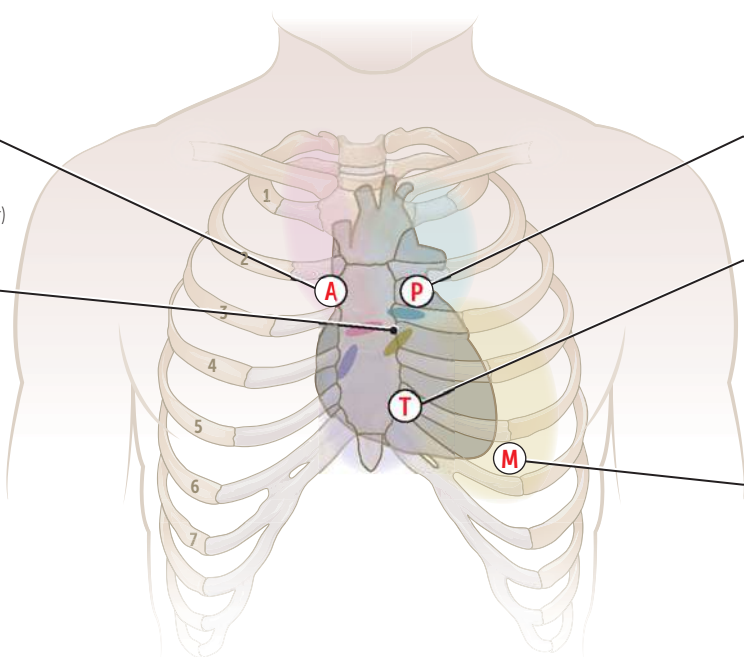
Where to listen: **APT M****Aortic area:****Systolic murmur**

Aortic stenosis
Flow murmur
(eg, physiologic murmur)
Aortic valve sclerosis

Left sternal border:

Diastolic murmur
Aortic regurgitation
Pulmonic regurgitation
Systolic murmur
Hypertrophic
cardiomyopathy

 Aortic
 Pulmonic
 Tricuspid
 Mitral

**Pulmonic area:**

Systolic ejection murmur
Pulmonic stenosis
Flow murmur

Tricuspid area:

Holosystolic murmur
Tricuspid regurgitation
Ventricular septal defect
Diastolic murmur
Tricuspid stenosis
Atrial septal defect (↑ flow
across tricuspid valve)

Mitral area (apex):

Holosystolic murmur
Mitral regurgitation
Systolic murmur
Mitral valve prolapse
Diastolic murmur
Mitral stenosis

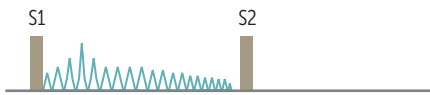


BEDSIDE MANEUVER	EFFECT
Inspiration (↑ venous return to right atrium)	↑ intensity of right heart sounds
Hand grip (↑ afterload)	↑ intensity of MR, AR, and VSD murmurs ↓ hypertrophic cardiomyopathy and AS murmurs MVP: later onset of click/murmur
Valsalva (phase II), standing up (↓ preload)	↓ intensity of most murmurs (including AS) ↑ intensity of hypertrophic cardiomyopathy murmur MVP: earlier onset of click/murmur
Rapid squatting (↑ venous return, ↑ preload, ↑ afterload)	↓ intensity of hypertrophic cardiomyopathy murmur ↑ intensity of AS, MR, and VSD murmurs MVP: later onset of click/murmur
Systolic heart sounds include the murmurs of aortic/pulmonic stenosis, mitral/tricuspid regurgitation, VSD, MVP, hypertrophic cardiomyopathy.	
Diastolic heart sounds include the murmurs of aortic/pulmonic regurgitation, mitral/tricuspid stenosis.	

Heart murmurs

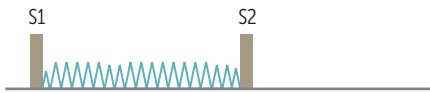
Systolic

Aortic stenosis



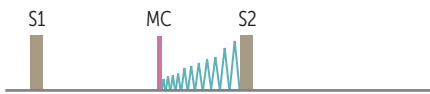
Crescendo-decrescendo systolic ejection murmur (ejection click may be present). LV $\gg$ aortic pressure during systole. Loudest at heart base; radiates to carotids. “Pulsus parvus et tardus”—pulses are weak with a delayed peak. Can lead to **S**yncope, **A**ngina, and **D**yspnea on exertion (**SAD**). Most commonly due to age-related calcification in older patients (> 60 years old) or in younger patients with early-onset calcification of bicuspid aortic valve.

Mitral/tricuspid regurgitation



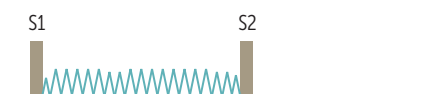
Holosystolic, high-pitched “blowing murmur.” Mitral—loudest at apex and radiates toward axilla. MR is often due to ischemic heart disease (post-MI), MVP, LV dilatation. Tricuspid—loudest at tricuspid area. TR commonly caused by RV dilatation. Rheumatic fever and infective endocarditis can cause either MR or TR.

Mitral valve prolapse



Late systolic crescendo murmur with midsystolic click (MC; due to sudden tensing of chordae tendineae). Most frequent valvular lesion. Best heard over apex. Loudest just before S2. Usually benign. Can predispose to infective endocarditis. Can be caused by myxomatous degeneration (1° or 2° to connective tissue disease such as Marfan or Ehlers-Danlos syndrome), rheumatic fever, chordae rupture.

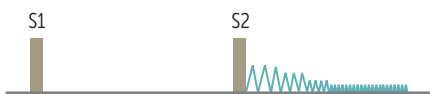
Ventricular septal defect



Holosystolic, harsh-sounding murmur. Loudest at tricuspid area.

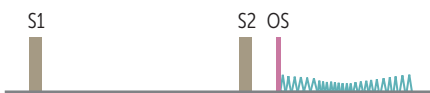
Diastolic

Aortic regurgitation



High-pitched “blowing” early diastolic decrescendo murmur. Long diastolic murmur, hyperdynamic pulse, and head bobbing when severe and chronic. Wide pulse pressure. Often due to aortic root dilation, bicuspid aortic valve, endocarditis, rheumatic fever. Progresses to left HF.

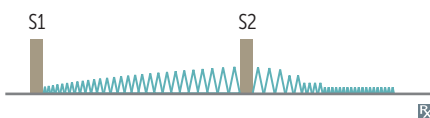
Mitral stenosis



Follows opening snap (OS; due to abrupt halt in leaflet motion in diastole, after rapid opening due to fusion at leaflet tips). Delayed rumbling mid-to-late diastolic murmur ($\downarrow$ interval between S2 and OS correlates with $\uparrow$ severity). LA $\gg$ LV pressure during diastole. Often a late (and highly specific) sequela of rheumatic fever. Chronic MS can result in LA dilatation.

Continuous

Patent ductus arteriosus



Continuous machine-like murmur. Loudest at S2. Often due to congenital rubella or prematurity. Best heard at left infraclavicular area.

Myocardial action potential

Also occurs in bundle of His and Purkinje fibers.

Phase 0 = rapid upstroke and depolarization—voltage-gated Na^+ channels open.

Phase 1 = initial repolarization—inactivation of voltage-gated Na^+ channels. Voltage-gated K^+ channels begin to open.

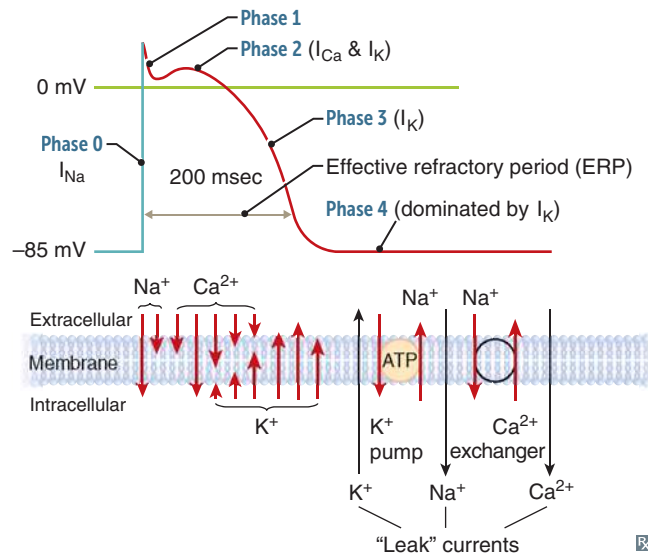
Phase 2 = plateau— Ca^{2+} influx through voltage-gated Ca^{2+} channels balances K^+ efflux. Ca^{2+} influx triggers Ca^{2+} release from sarcoplasmic reticulum and myocyte contraction.

Phase 3 = rapid repolarization—massive K^+ efflux due to opening of voltage-gated slow K^+ channels and closure of voltage-gated Ca^{2+} channels.

Phase 4 = resting potential—high K^+ permeability through K^+ channels.

In contrast to skeletal muscle:

- Cardiac muscle action potential has a plateau, which is due to Ca^{2+} influx and K^+ efflux.
- Cardiac muscle contraction requires Ca^{2+} influx from ECF to induce Ca^{2+} release from sarcoplasmic reticulum (Ca^{2+} -induced Ca^{2+} release).
- Cardiac myocytes are electrically coupled to each other by gap junctions.



Pacemaker action potential

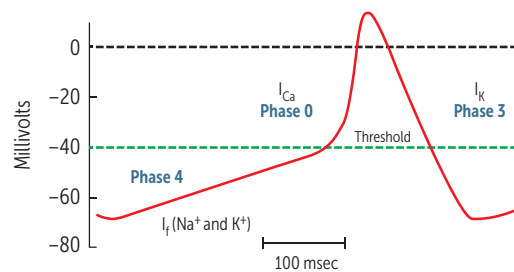
Occurs in the SA and AV nodes. Key differences from the ventricular action potential include:

Phase 0 = upstroke—opening of voltage-gated Ca^{2+} channels. Fast voltage-gated Na^+ channels are permanently inactivated because of the less negative resting potential of these cells. Results in a slow conduction velocity that is used by the AV node to prolong transmission from the atria to ventricles.

Phases 1 and 2 are absent.

Phase 3 = inactivation of the Ca^{2+} channels and $\uparrow$ activation of K^+ channels $\rightarrow \uparrow \text{K}^+$ efflux.

Phase 4 = slow spontaneous diastolic depolarization due to I_f (“funny current”). I_f channels responsible for a slow, mixed Na^+/K^+ inward current; different from I_{Na} in phase 0 of ventricular action potential. Accounts for automaticity of SA and AV nodes. The slope of phase 4 in the SA node determines HR. ACh/adenosine $\downarrow$ the rate of diastolic depolarization and $\downarrow$ HR, while catecholamines $\uparrow$ depolarization and $\uparrow$ HR. Sympathetic stimulation $\uparrow$ the chance that I_f channels are open and thus $\uparrow$ HR.



Electrocardiogram

Conduction pathway—SA node → atria
→ AV node → bundle of His → right and
left bundle branches → Purkinje fibers
→ ventricles; left bundle branch divides into
left anterior and posterior fascicles.

SA node “pacemaker” inherent dominance with
slow phase of upstroke.

AV node—located in posteroinferior part of
interatrial septum. Blood supply usually
from RCA. 100-msec delay allows time for
ventricular filling.

Pacemaker rates—SA > AV > bundle of His/
Purkinje/ventricles.

Speed of conduction—Purkinje > atria
> ventricles > AV node.

P wave—atrial depolarization. Atrial
repolarization is masked by QRS complex.

PR interval—time from start of atrial
depolarization to start of ventricular
depolarization (normally < 200 msec).

QRS complex—ventricular depolarization
(normally < 120 msec).

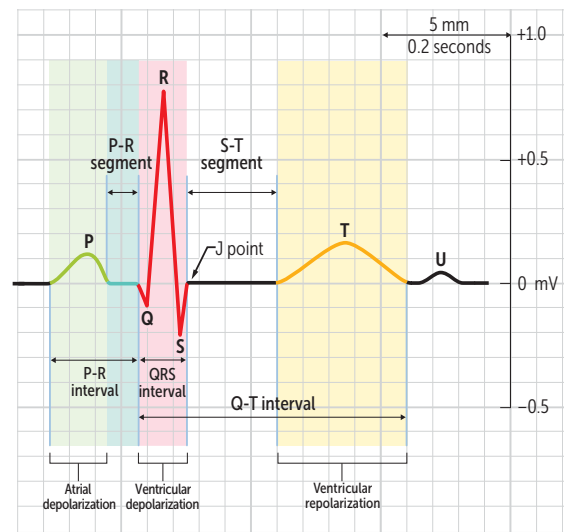
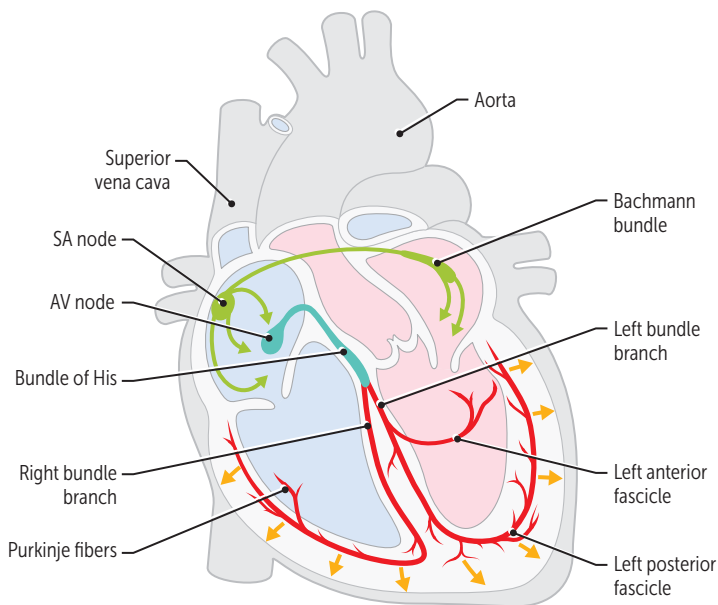
QT interval—ventricular depolarization,
mechanical contraction of the ventricles,
ventricular repolarization.

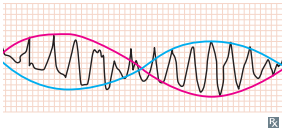
T wave—ventricular repolarization. T-wave
inversion may indicate ischemia or recent MI.

J point—junction between end of QRS complex
and start of ST segment.

ST segment—isolectric, ventricles depolarized.

U wave—prominent in hypokalemia,
bradycardia.



Torsades de pointes

Polymorphic ventricular tachycardia, characterized by shifting sinusoidal waveforms on ECG; can progress to ventricular fibrillation (VF). Long QT interval predisposes to torsades de pointes. Caused by drugs, ↓ K⁺, ↓ Mg²⁺, congenital abnormalities. Treatment includes magnesium sulfate.

Drug-induced long QT (**ABCDE**):

- Anti**A**rrhythmics (class IA, III)
- Anti**B**iotics (eg, macrolides)
- Anti“**C**”ychotics (eg, haloperidol)
- Anti**D**epressants (eg, TCAs)
- Anti**E**metics (eg, ondansetron)

Torsades de pointes = twisting of the points

Congenital long QT syndrome

Inherited disorder of myocardial repolarization, typically due to ion channel defects; ↑ risk of sudden cardiac death (SCD) due to torsades de pointes. Includes:

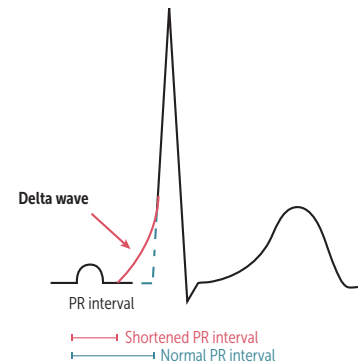
- **Romano-Ward syndrome**—autosomal dominant, pure cardiac phenotype (no deafness).
- **Jervell and Lange-Nielsen syndrome**—autosomal recessive, sensorineural deafness.

Brugada syndrome

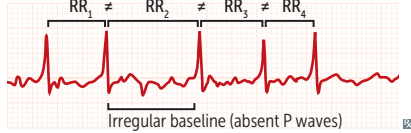
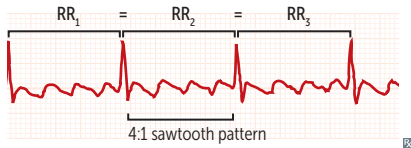
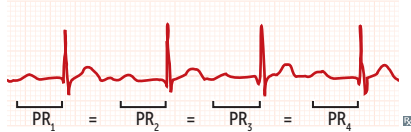
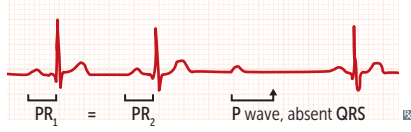
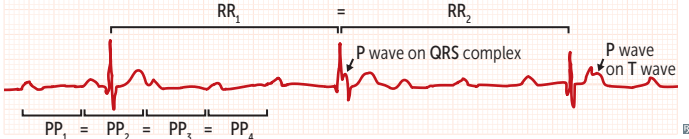
Autosomal dominant disorder most common in Asian males. ECG pattern of pseudo-right bundle branch block and ST elevations in V₁-V₃. ↑ risk of ventricular tachyarrhythmias and SCD. Prevent SCD with implantable cardioverter-defibrillator (ICD).

Wolff-Parkinson-White syndrome

Most common type of ventricular pre-excitation syndrome. Abnormal fast accessory conduction pathway from atria to ventricle (bundle of Kent) bypasses the rate-slowing AV node → ventricles begin to partially depolarize earlier → characteristic delta wave with widened QRS complex and shortened PR interval on ECG. May result in reentry circuit → supraventricular tachycardia.



ECG tracings

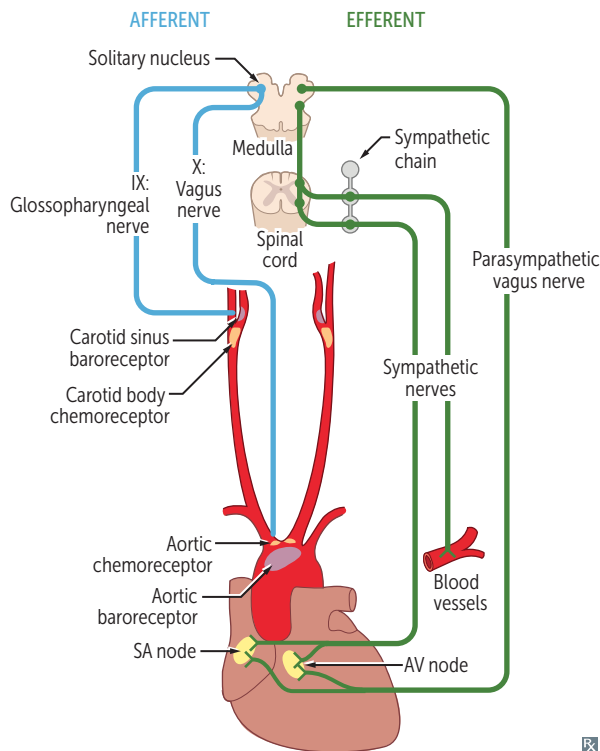
RHYTHM	DESCRIPTION	EXAMPLE
Atrial fibrillation	Chaotic and erratic baseline with no discrete P waves in between irregularly spaced QRS complexes. Irregularly irregular heartbeat. Most common risk factors include hypertension and coronary artery disease (CAD). Can lead to thromboembolic events, particularly stroke. Treatment includes anticoagulation, rate control, rhythm control, and/or cardioversion.	
Atrial flutter	A rapid succession of identical, back-to-back atrial depolarization waves. The identical appearance accounts for the “sawtooth” appearance of the flutter waves. Treat like atrial fibrillation. Definitive treatment is catheter ablation.	
Ventricular fibrillation	A completely erratic rhythm with no identifiable waves. Fatal arrhythmia without immediate CPR and defibrillation.	
AV block		
First degree	The PR interval is prolonged (> 200 msec). Benign and asymptomatic. No treatment required.	
Second degree		
Mobitz type I (Wenckebach)	Progressive lengthening of PR interval until a beat is “dropped” (a P wave not followed by a QRS complex). Usually asymptomatic. Variable RR interval with a pattern (regularly irregular).	
Mobitz type II	Dropped beats that are not preceded by a change in the length of the PR interval (as in type I). May progress to 3rd-degree block. Often treated with pacemaker.	
Third degree (complete)	The atria and ventricles beat independently of each other. P waves and QRS complexes not rhythmically associated. Atrial rate > ventricular rate. Usually treated with pacemaker. Can be caused by Lyme disease.	

Atrial natriuretic peptide

Released from **atrial myocytes** in response to $\uparrow$ blood volume and atrial pressure. Acts via cGMP. Causes vasodilation and $\downarrow$ Na^+ reabsorption at the renal collecting tubule. Dilates afferent renal arterioles and constricts efferent arterioles, promoting diuresis and contributing to “aldosterone escape” mechanism.

B-type (brain) natriuretic peptide

Released from **ventricular myocytes** in response to $\uparrow$ tension. Similar physiologic action to ANP, with longer half-life. BNP blood test used for diagnosing HF (very good negative predictive value). Available in recombinant form (nesiritide) for treatment of HF.

Baroreceptors and chemoreceptors**Receptors:**

- Aortic arch transmits via vagus nerve to solitary nucleus of medulla (responds to $\downarrow$ and $\uparrow$ in BP).
- Carotid sinus (dilated region at carotid bifurcation) transmits via glossopharyngeal nerve to solitary nucleus of medulla (responds to $\downarrow$ and $\uparrow$ in BP).

Baroreceptors:

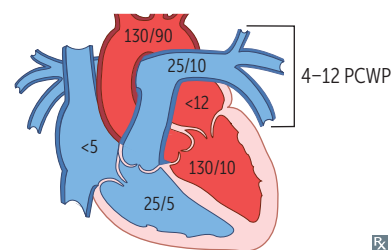
- Hypotension— $\downarrow$ arterial pressure $\rightarrow$ $\downarrow$ stretch $\rightarrow$ $\downarrow$ afferent baroreceptor firing $\rightarrow$ $\uparrow$ efferent sympathetic firing and $\downarrow$ efferent parasympathetic stimulation $\rightarrow$ vasoconstriction, $\uparrow$ HR, $\uparrow$ contractility, $\uparrow$ BP. Important in the response to severe hemorrhage.
- Carotid massage— $\uparrow$ pressure on carotid sinus $\rightarrow$ $\uparrow$ stretch $\rightarrow$ $\uparrow$ afferent baroreceptor firing $\rightarrow$ $\uparrow$ AV node refractory period $\rightarrow$ $\downarrow$ HR.
- Component of Cushing reflex (triad of hypertension, bradycardia, and respiratory depression)— $\uparrow$ intracranial pressure constricts arterioles $\rightarrow$ cerebral ischemia $\rightarrow$ $\uparrow$ pCO_2 and $\downarrow$ pH $\rightarrow$ central reflex sympathetic $\uparrow$ in perfusion pressure (hypertension) $\rightarrow$ $\uparrow$ stretch $\rightarrow$ peripheral reflex baroreceptor-induced bradycardia.

Chemoreceptors:

- Peripheral—carotid and aortic bodies are stimulated by $\downarrow$ PO_2 (< 60 mm Hg), $\uparrow$ PCO_2 , and $\downarrow$ pH of blood.
- Central—are stimulated by changes in pH and PCO_2 of brain interstitial fluid, which in turn are influenced by arterial CO_2 . Do not directly respond to PO_2 .

Normal cardiac pressures

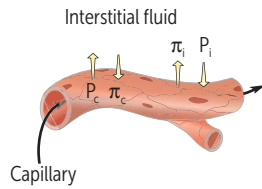
Pulmonary capillary wedge pressure (PCWP; in mm Hg) is a good approximation of left atrial pressure. In mitral stenosis, PCWP > LV end diastolic pressure. PCWP is measured with pulmonary artery catheter (Swan-Ganz catheter).

**Autoregulation**

How blood flow to an organ remains constant over a wide range of perfusion pressures.

ORGAN	FACTORS DETERMINING AUTOREGULATION	
Heart	Local metabolites (vasodilatory): adenosine, NO, CO ₂ , ↓ O ₂	Note: the pulmonary vasculature is unique in that hypoxia causes vasoconstriction so that only well-ventilated areas are perfused. In other organs, hypoxia causes vasodilation.
Brain	Local metabolites (vasodilatory): CO ₂ (pH)	
Kidneys	Myogenic and tubuloglomerular feedback	
Lungs	Hypoxia causes vasoconstriction	
Skeletal muscle	Local metabolites during exercise: lactate, adenosine, K ⁺ , H ⁺ , CO ₂ At rest: sympathetic tone	CO ₂ , H ⁺ , Adenosine, Lactate, K ⁺ (CHALK)
Skin	Sympathetic stimulation most important mechanism for temperature control	

Capillary fluid exchange



Starling forces determine fluid movement through capillary membranes:

- P_c = capillary pressure—pushes fluid out of capillary
- P_i = interstitial fluid pressure—pushes fluid into capillary
- π_c = plasma colloid osmotic (oncotic) pressure—pulls fluid into capillary
- π_i = interstitial fluid colloid osmotic pressure—pulls fluid out of capillary

$$J_v = \text{net fluid flow} = K_f [(P_c - P_i) - \zeta(\pi_c - \pi_i)]$$

K_f = capillary permeability to fluid

ζ = reflection coefficient (measure of capillary permeability to protein)

Edema—excess fluid outflow into interstitium commonly caused by:

- $\uparrow$ capillary pressure ($\uparrow P_c$; eg, HF)
- $\downarrow$ plasma proteins ($\downarrow \pi_c$; eg, nephrotic syndrome, liver failure, protein malnutrition)
- $\uparrow$ capillary permeability ($\uparrow K_f$; eg, toxins, infections, burns)
- $\uparrow$ interstitial fluid colloid osmotic pressure ($\uparrow \pi_i$; eg, lymphatic blockage)

► CARDIOVASCULAR—PATHOLOGY

Congenital heart diseases

RIGHT-TO-LEFT SHUNTS

Early cyanosis—“blue babies.” Often diagnosed prenatally or become evident immediately after birth. Usually require urgent surgical treatment and/or maintenance of a PDA.

The **5 Ts**:

1. **T**runcus arteriosus (**1** vessel)
2. **T**ransposition (**2** switched vessels)
3. **T**ricuspid atresia (**3** = **Tri**)
4. **T**etralogy of Fallot (**4** = **Tetra**)
5. **TAPVR** (**5** letters in the name)

Persistent truncus arteriosus

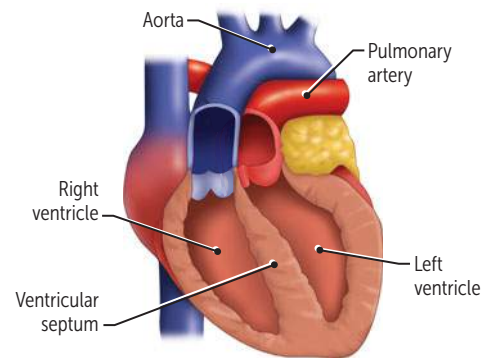
Truncus arteriosus fails to divide into pulmonary trunk and aorta due to lack of aorticopulmonary septum formation; most patients have accompanying VSD.

D-transposition of great vessels

Aorta leaves RV (anterior) and pulmonary trunk leaves LV (posterior) → separation of systemic and pulmonary circulations. Not compatible with life unless a shunt is present to allow mixing of blood (eg, VSD, PDA, or patent foramen ovale).

Due to failure of the aorticopulmonary septum to spiral.

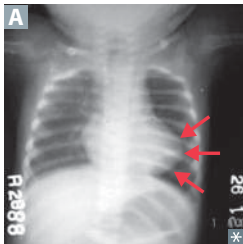
Without surgical intervention, most infants die within the first few months of life.



Tricuspid atresia

Absence of tricuspid valve and hypoplastic RV; requires both ASD and VSD for viability.

Tetralogy of Fallot



Caused by anterosuperior displacement of the infundibular septum. Most common cause of early childhood cyanosis.

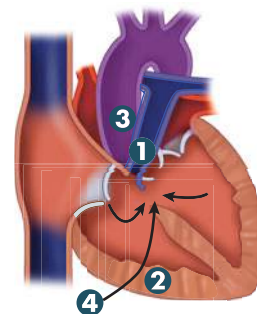
- 1 **P**ulmonary infundibular stenosis (most important determinant for prognosis)
- 2 **R**ight ventricular hypertrophy (RVH)—boot-shaped heart on CXR **A**
- 3 **O**verriding aorta
- 4 **V**SD

Pulmonary stenosis forces right-to-left flow across VSD → RVH, “tet spells” (often caused by crying, fever, and exercise due to exacerbation of RV outflow obstruction).

PROVe.

Squatting: ↑ SVR, ↓ right-to-left shunt, improves cyanosis.

Treatment: early surgical correction.



Total anomalous pulmonary venous return

Pulmonary veins drain into right heart circulation (SVC, coronary sinus, etc); associated with ASD and sometimes PDA to allow for right-to-left shunting to maintain CO.

Ebstein anomaly

Characterized by displacement of tricuspid valve leaflets downward into RV, artificially “atrializing” the ventricle. Associated with tricuspid regurgitation and right HF. Can be caused by lithium exposure in utero.

Congenital heart diseases (continued)**LEFT-TO-RIGHT SHUNTS**

Acyanotic at presentation; cyanosis may occur years later.

Frequency: VSD > ASD > PDA.

Right-to-Left shunts: ea**RL**y cyanosis.

Left-to-Right shunts: “**LateR**” cyanosis.

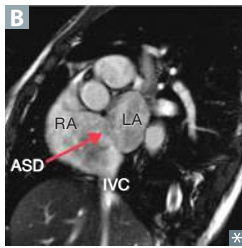
Ventricular septal defect

Most common congenital cardiac defect.

Asymptomatic at birth, may manifest weeks later or remain asymptomatic throughout life.

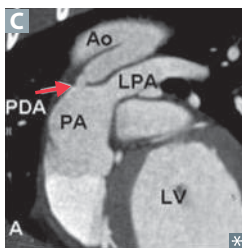
Most self resolve; larger lesions may lead to LV overload and HF.

O₂ saturation ↑ in RV and pulmonary artery.

Atrial septal defect

Defect in interatrial septum **B**; loud S1; wide, fixed split S2. Ostium secundum defects most common and usually occur as isolated findings; ostium primum defects rarer yet usually occur with other cardiac anomalies. Symptoms range from none to HF. Distinct from patent foramen ovale in that septa are missing tissue rather than unfused.

O₂ saturation ↑ in RA, RV, and pulmonary artery. May lead to paradoxical emboli (systemic venous emboli use ASD to bypass lungs and become systemic arterial emboli).

Patent ductus arteriosus

In fetal period, shunt is right to left (normal).

In neonatal period, ↓ pulmonary vascular resistance → shunt becomes left to right → progressive RVH and/or LVH and HF.

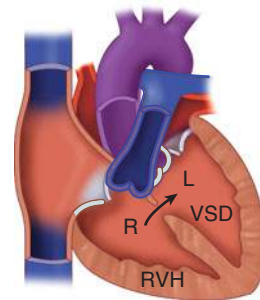
Associated with a continuous, “machine-like” murmur. Patency is maintained by PGE synthesis and low O₂ tension. Uncorrected PDA **C** can eventually result in late cyanosis in the lower extremities (differential cyanosis).

“**Endomethacin**” (indomethacin) **ends** patency of PDA; PGE keeps ductus **G**oing (may be necessary to sustain life in conditions such as transposition of the great vessels).

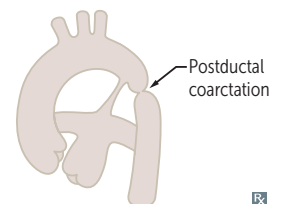
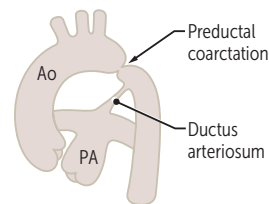
PDA is normal in utero and normally closes only after birth.

Eisenmenger syndrome

Uncorrected left-to-right shunt (VSD, ASD, PDA) → ↑ pulmonary blood flow → pathologic remodeling of vasculature → pulmonary arterial hypertension. RVH occurs to compensate → shunt becomes right to left. Causes late cyanosis, clubbing **D**, and polycythemia. Age of onset varies.

**OTHER ANOMALIES****Coarctation of the aorta**

Aortic narrowing near insertion of ductus arteriosus (“juxtaductal”). Associated with bicuspid aortic valve, other heart defects, and Turner syndrome. Hypertension in upper extremities and weak, delayed pulse in lower extremities (brachial-femoral delay). With age, intercostal arteries enlarge due to collateral circulation; arteries erode ribs → notched appearance on CXR. Complications include HF, ↑ risk of cerebral hemorrhage (berry aneurysms), aortic rupture, and possible endocarditis.



Congenital cardiac defect associations

DISORDER	DEFECT
Alcohol exposure in utero (fetal alcohol syndrome)	VSD, PDA, ASD, tetralogy of Fallot
Congenital rubella	PDA, pulmonary artery stenosis, septal defects
Down syndrome	AV septal defect (endocardial cushion defect), VSD, ASD
Infant of diabetic mother	Transposition of great vessels
Marfan syndrome	MVP, thoracic aortic aneurysm and dissection, aortic regurgitation
Prenatal lithium exposure	Ebstein anomaly
Turner syndrome	Bicuspid aortic valve, coarctation of aorta
Williams syndrome	Supravalvular aortic stenosis
22q11 syndromes	Truncus arteriosus, tetralogy of Fallot

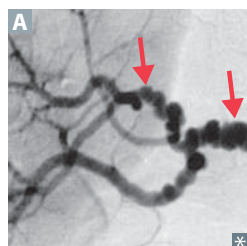
Hypertension

Defined as persistent systolic BP ≥ 140 mm Hg and/or diastolic BP ≥ 90 mm Hg

RISK FACTORS

↑ age, obesity, diabetes, physical inactivity, excess salt intake, excess alcohol intake, family history; African American > Caucasian > Asian.

FEATURES



90% of hypertension is 1° (essential) and related to ↑ CO or ↑ TPR. Remaining 10% mostly 2° to renal/renovascular disease (eg, fibromuscular dysplasia [which has characteristic “string of beads” appearance of renal artery **A**], atherosclerosis) and 1° hyperaldosteronism.

Hypertensive urgency—severe ($\geq 180/\geq 120$ mm Hg) hypertension without acute end-organ damage.

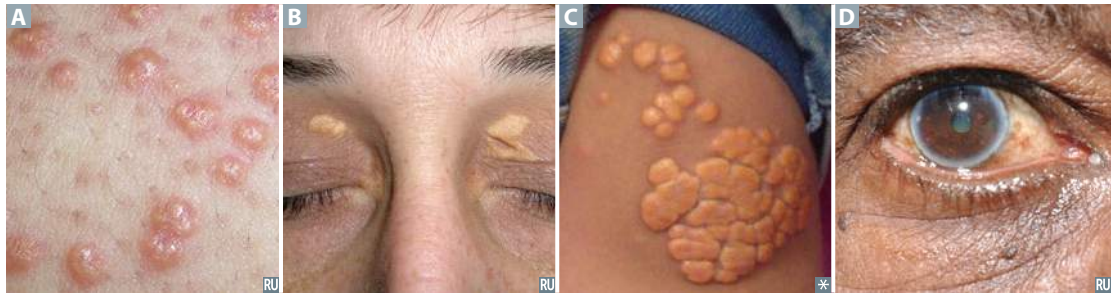
Hypertensive emergency—severe hypertension with evidence of acute end-organ damage (eg, encephalopathy, stroke, retinal hemorrhages and exudates, papilledema, MI, HF, aortic dissection, kidney injury, microangiopathic hemolytic anemia, eclampsia).

PREDISPOSES TO

CAD, LVH, HF, atrial fibrillation; aortic dissection, aortic aneurysm; stroke; chronic kidney disease (hypertensive nephropathy); retinopathy.

Hyperlipidemia signs

Xanthomas	Plaques or nodules composed of lipid-laden histiocytes in skin A , especially the eyelids (xanthelasma B).
Tendinous xanthoma	Lipid deposit in tendon C , especially Achilles.
Corneal arcus	Lipid deposit in cornea. Common in elderly (arcus senilis D), but appears earlier in life in hypercholesterolemia.

**Arteriosclerosis**

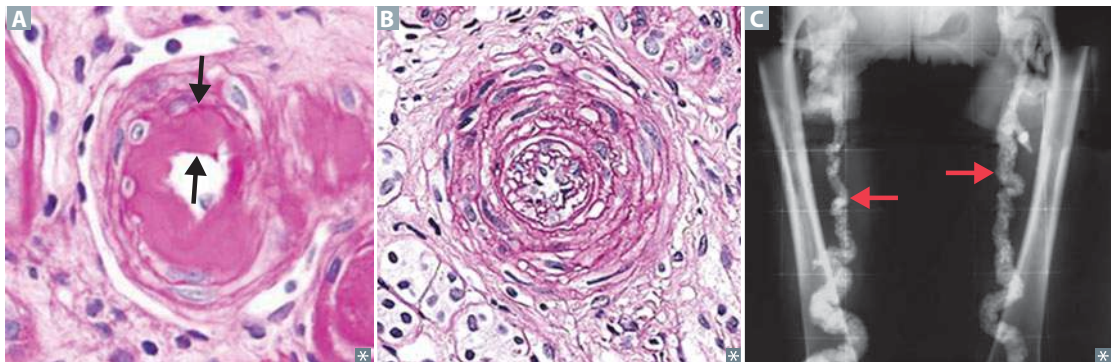
Hardening of arteries, with arterial wall thickening and loss of elasticity.

Arteriolosclerosis

Common. Affects small arteries and arterioles. Two types: hyaline (thickening of vessel walls in essential hypertension or diabetes mellitus **A**) and hyperplastic (“onion skinning” in severe hypertension **B** with proliferation of smooth muscle cells).

Mönckeberg sclerosis (medial calcific sclerosis)

Uncommon. Affects medium-sized arteries. Calcification of internal elastic lamina and media of arteries → vascular stiffening without obstruction. “Pipestem” appearance on x-ray **C**. Does not obstruct blood flow; intima not involved.

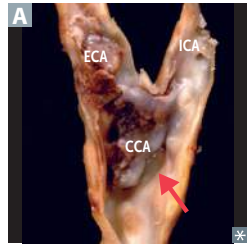


Atherosclerosis

Very common. Disease of elastic arteries and large- and medium-sized muscular arteries; a form of arteriosclerosis caused by buildup of cholesterol plaques.

LOCATION

Abdominal aorta > coronary artery > popliteal artery > carotid artery **A**.

**RISK FACTORS**

Modifiable: smoking, hypertension, hyperlipidemia (↑ LDL), diabetes.

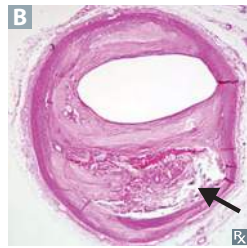
Nonmodifiable: age, sex (↑ in men and postmenopausal women), family history.

SYMPTOMS

Angina, claudication, but can be asymptomatic.

PROGRESSION

Inflammation important in pathogenesis: endothelial cell dysfunction → macrophage and LDL accumulation → foam cell formation → fatty streaks → smooth muscle cell migration (involves PDGF and FGF), proliferation, and extracellular matrix deposition → fibrous plaque → complex atheromas **B**.

**COMPLICATIONS**

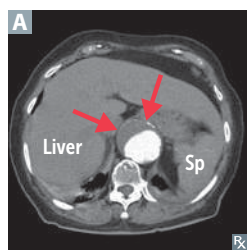
Aneurysms, ischemia, infarcts, peripheral vascular disease, thrombus, emboli.

Aortic aneurysm

Localized pathologic dilatation of the aorta. May cause abdominal and/or back pain, which is a sign of leaking, dissection, or imminent rupture.

Abdominal aortic aneurysm

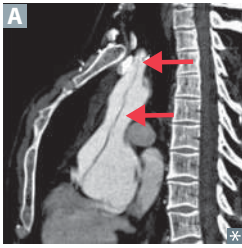
Associated with atherosclerosis. Risk factors include history of tobacco use, ↑ age, male sex, family history. May present as palpable pulsatile abdominal mass (arrows in **A** point to outer dilated calcified aortic wall, with partial crescent-shaped nonopacification of aorta due to flap/clot).

**Thoracic aortic aneurysm**

Associated with cystic medial degeneration. Risk factors include hypertension, bicuspid aortic valve, connective tissue disease (eg, Marfan syndrome). Also historically associated with 3° syphilis (obliterative endarteritis of the vasa vasorum). Aortic root dilatation may lead to aortic valve regurgitation.

Traumatic aortic rupture

Due to trauma and/or deceleration injury, most commonly at aortic isthmus (proximal descending aorta just distal to origin of left subclavian artery).

Aortic dissection

Longitudinal intimal tear forming a false lumen (arrows in **A** show flap extending into ascending aorta, Stanford type A dissection). Associated with hypertension, bicuspid aortic valve, inherited connective tissue disorders (eg, Marfan syndrome). Can present with tearing, sudden-onset chest pain, radiating to the back +/- markedly unequal BP in arms. CXR shows mediastinal widening. Can result in organ ischemia, aortic rupture, death. Two types:

- Stanford type **A** (proximal): involves **A**scending aorta. May extend to aortic arch or descending aorta. May result in acute aortic regurgitation or cardiac tamponade. Treatment: surgery.
- Stanford type **B** (distal): only involves descending aorta (**B**elow ligamentum arteriosum). Treat medically with β -blockers, then vasodilators.

Ischemic heart disease manifestations**Angina**

Chest pain due to ischemic myocardium 2° to coronary artery narrowing or spasm; no myocyte necrosis.

- **Stable**—usually 2° to atherosclerosis; exertional chest pain in classic distribution (usually with ST depression on ECG), resolving with rest or nitroglycerin.
- **Variant (Prinzmetal)**—occurs at rest 2° to coronary artery spasm; transient ST elevation on ECG. Smoking is a risk factor, but hypertension and hypercholesterolemia are not. Triggers may include cocaine, alcohol, and triptans. Treat with Ca^{2+} channel blockers, nitrates, and smoking cessation (if applicable).
- **Unstable**—thrombosis with incomplete coronary artery occlusion; +/- ST depression and/or T-wave inversion on ECG but no cardiac biomarker elevation (unlike NSTEMI); ↑ in frequency or intensity of chest pain or any chest pain at rest.

Coronary steal syndrome

Distal to coronary stenosis, vessels are maximally dilated at baseline. Administration of vasodilators (eg, dipyridamole, regadenoson) dilates normal vessels and shunts blood toward well-perfused areas, thereby diverting flow away from vessels that are stenosed and leading to ischemia in myocardium perfused by these vessels. Principle behind pharmacologic stress tests with coronary vasodilators.

Sudden cardiac death

Death from cardiac causes within 1 hour of onset of symptoms, most commonly due to a lethal arrhythmia (eg, VF). Associated with CAD (up to 70% of cases), cardiomyopathy (hypertrophic, dilated), and hereditary ion channelopathies (eg, long QT syndrome, Brugada syndrome). Prevent with implantable cardioverter-defibrillator (ICD).

Chronic ischemic heart disease

Progressive onset of HF over many years due to chronic ischemic myocardial damage.

Myocardial infarction

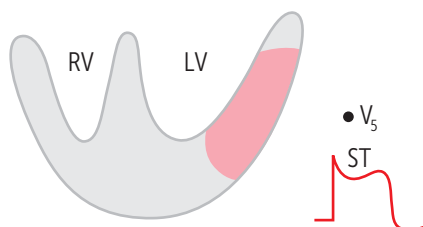
Most often acute thrombosis due to rupture of coronary artery atherosclerotic plaque. ↑ cardiac biomarkers (CK-MB, troponins) are diagnostic.

ST-segment elevation MI (STEMI)

Transmural infarcts

Full thickness of myocardial wall involved

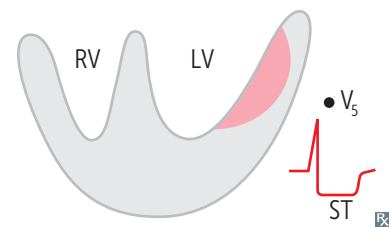
ST elevation on ECG, Q waves

**Non-ST-segment elevation MI (NSTEMI)**

Subendocardial infarcts

Subendocardium (inner 1/3) especially vulnerable to ischemia

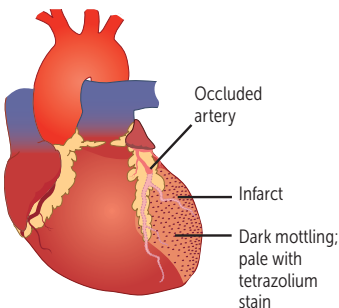
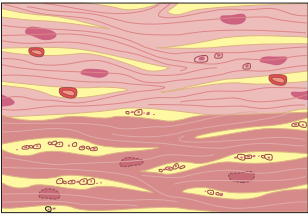
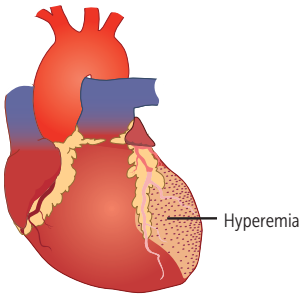
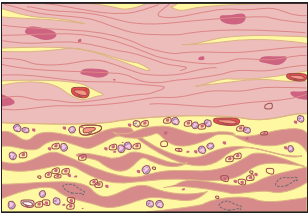
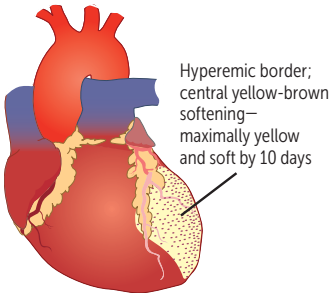
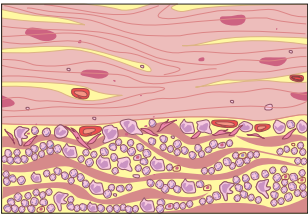
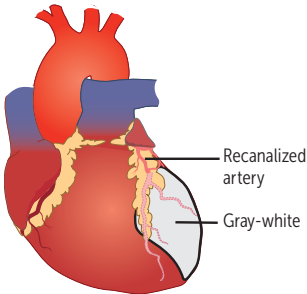
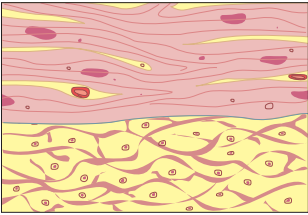
ST depression on ECG



Evolution of myocardial infarction

Commonly occluded coronary arteries: LAD > RCA > circumflex.

Symptoms: diaphoresis, nausea, vomiting, severe retrosternal pain, pain in left arm and/or jaw, shortness of breath, fatigue.

TIME	GROSS	LIGHT MICROSCOPE	COMPLICATIONS
0–24 hr	None 	Early coagulative necrosis, release of necrotic cell contents into blood; edema, hemorrhage, wavy fibers. Neutrophils appear. Reperfusion injury, associated with generation of free radicals, leads to hypercontraction of myofibrils through ↑ free calcium influx. 	Ventricular arrhythmia, HF, cardiogenic shock.
1–3 days		Extensive coagulative necrosis. Tissue surrounding infarct shows acute inflammation with neutrophils. 	Postinfarction fibrinous pericarditis.
3–14 days		Macrophages, then granulation tissue at margins. 	Free wall rupture → tamponade; papillary muscle rupture → mitral regurgitation; interventricular septal rupture due to macrophage-mediated structural degradation. LV pseudoaneurysm (risk of rupture).
2 weeks to several months		Contracted scar complete. 	Dressler syndrome, HF, arrhythmias, true ventricular aneurysm (risk of mural thrombus).

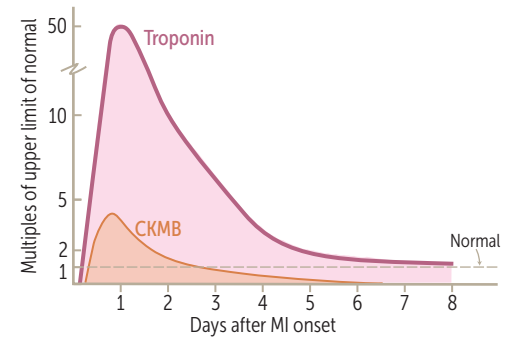
Diagnosis of myocardial infarction

In the first 6 hours, ECG is the gold standard. Cardiac troponin I rises after 4 hours (peaks at 24 hr) and is ↑ for 7–10 days; more specific than other protein markers.

CK-MB rises after 6–12 hours (peaks at 16–24 hr) and is predominantly found in myocardium but can also be released from skeletal muscle. Useful in diagnosing reinfarction following acute MI because levels return to normal after 48 hours.

Large MIs lead to greater elevations in troponin I and CK-MB. Exact curves vary with testing procedure.

ECG changes can include ST elevation (STEMI, transmural infarct), ST depression (NSTEMI, subendocardial infarct), hyperacute (peaked) T waves, T-wave inversion, new left bundle branch block, and pathologic Q waves or poor R wave progression (evolving or old transmural infarct).

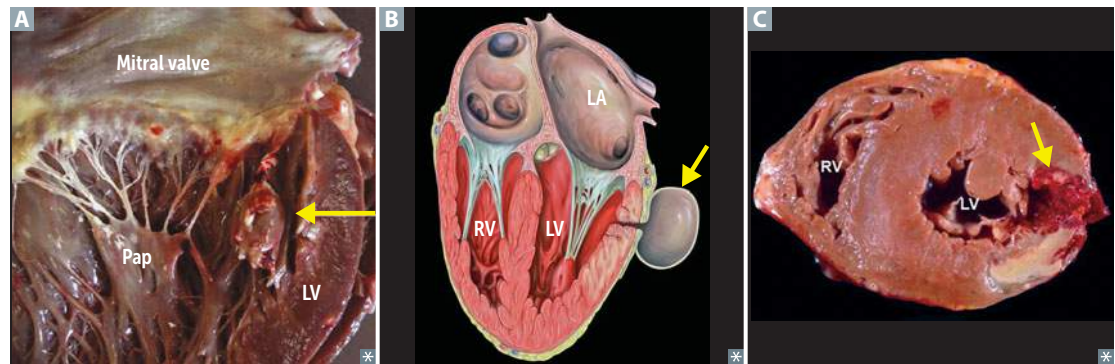


ECG localization of STEMI

INFARCT LOCATION	LEADS WITH ST ELEVATIONS OR Q WAVES
Anteroseptal (LAD)	V ₁ –V ₂
Anteroapical (distal LAD)	V ₃ –V ₄
Anterolateral (LAD or LCX)	V ₅ –V ₆
Lateral (LCX)	I, aVL
Inferior (RCA)	II, III, aVF
Posterior (PDA)	V ₇ –V ₉ , ST depression in V ₁ –V ₃ with tall R waves

Myocardial infarction complications

Cardiac arrhythmia	Occurs within the first few days after MI. Important cause of death before reaching the hospital and within the first 24 hours post-MI.
Postinfarction fibrinous pericarditis	Occurs 1–3 days after MI. Friction rub.
Papillary muscle rupture	Occurs 2–7 days after MI. Posteromedial papillary muscle rupture A ↑ risk due to single blood supply from posterior descending artery. Can result in severe mitral regurgitation.
Interventricular septal rupture	Occurs 3–5 days after MI. Macrophage-mediated degradation → VSD.
Ventricular pseudoaneurysm formation	Occurs 3–14 days after MI. Contained free wall rupture B ; ↓ CO, risk of arrhythmia, embolus from mural thrombus.
Ventricular free wall rupture	Occurs 5–14 days after MI. Free wall rupture C → cardiac tamponade. LV hypertrophy and previous MI protect against free wall rupture.
True ventricular aneurysm	Occurs 2 weeks to several months after MI. Outward bulge with contraction (“dyskinesia”), associated with fibrosis.
Dressler syndrome	Occurs several weeks after MI. Autoimmune phenomenon resulting in fibrinous pericarditis.
LV failure and pulmonary edema	Can occur 2° to LV infarction, VSD, free wall rupture, papillary muscle rupture with mitral regurgitation.

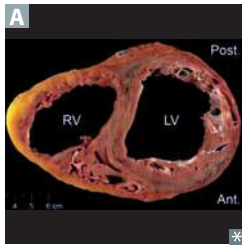
**Acute coronary syndrome treatments**

Unstable angina/NSTEMI—Anticoagulation (eg, heparin), antiplatelet therapy (eg, aspirin) + ADP receptor inhibitors (eg, clopidogrel), β -blockers, ACE inhibitors, statins. Symptom control with nitroglycerin and morphine.

STEMI—In addition to above, reperfusion therapy most important (percutaneous coronary intervention preferred over fibrinolysis).

Cardiomyopathies

Dilated cardiomyopathy



Most common cardiomyopathy (90% of cases). Often idiopathic or familial. Other etiologies include chronic **A**lcohol abuse, wet **B**eriberi, **C**oxsackie B viral myocarditis, chronic **C**ocaine use, **C**hagas disease, **D**oxorubicin toxicity, hemochromatosis, sarcoidosis, peripartum cardiomyopathy.

Findings: HF, S3, systolic regurgitant murmur, dilated heart on echocardiogram, balloon appearance of heart on CXR.

Treatment: Na⁺ restriction, ACE inhibitors, β-blockers, diuretics, digoxin, ICD, heart transplant.

Systolic dysfunction ensues.

Eccentric hypertrophy **A** (sarcomeres added in series).

ABCCCD.

Takotsubo cardiomyopathy: “broken heart syndrome”—ventricular apical ballooning likely due to increased sympathetic stimulation (stressful situations).

Hypertrophic cardiomyopathy



60–70% of cases are familial, autosomal dominant (most commonly due to mutations in genes encoding sarcomeric proteins, such as myosin binding protein C and β-myosin heavy chain). Can be associated with Friedreich ataxia. Causes syncope during exercise and may lead to sudden death in young athletes due to ventricular arrhythmia.

Findings: S4, systolic murmur. May see mitral regurgitation due to impaired mitral valve closure.

Treatment: cessation of high-intensity athletics, use of β-blocker or non-dihydropyridine Ca²⁺ channel blockers (eg, verapamil). ICD if patient is high risk.

Diastolic dysfunction ensues.

Marked ventricular concentric hypertrophy (sarcomeres added in parallel) **B**, often septal predominance. Myofibrillar disarray and fibrosis.

Hypertrophic obstructive cardiomyopathy (subset)—asymmetric septal hypertrophy and systolic anterior motion of mitral valve → outflow obstruction → dyspnea, possible syncope.

Restrictive/infiltrative cardiomyopathy

Postradiation fibrosis, **L**öffler syndrome, **E**ndocardial fibroelastosis (thick fibroelastic tissue in endocardium of young children), **A**myloidosis, **S**arcoidosis, **H**emochromatosis (although dilated cardiomyopathy is more common) (**P**uppy **LEASH**).

Diastolic dysfunction ensues. Can have low-voltage ECG despite thick myocardium (especially amyloid).

Löffler syndrome—endomyocardial fibrosis with a prominent eosinophilic infiltrate.

Heart failure

Clinical syndrome of cardiac pump dysfunction → congestion and low perfusion. Symptoms include dyspnea, orthopnea, fatigue; signs include S3 heart sound, rales, jugular venous distention (JVD), pitting edema **A**.

Systolic dysfunction—reduced EF, ↑ EDV; ↓ contractility often 2° to ischemia/MI or dilated cardiomyopathy.

Diastolic dysfunction—preserved EF, normal EDV; ↓ compliance often 2° to myocardial hypertrophy.

Right HF most often results from left HF. Cor pulmonale refers to isolated right HF due to pulmonary cause.

ACE inhibitors or angiotensin II receptor blockers, β-blockers (except in acute decompensated HF), and spironolactone ↓ mortality. Thiazide or loop diuretics are used mainly for symptomatic relief.

Hydralazine with nitrate therapy improves both symptoms and mortality in select patients.

Left heart failure**Orthopnea**

Shortness of breath when supine: ↑ venous return from redistribution of blood (immediate gravity effect) exacerbates pulmonary vascular congestion.

Paroxysmal nocturnal dyspnea

Breathless awakening from sleep: ↑ venous return from redistribution of blood, reabsorption of peripheral edema, etc.

Pulmonary edema

↑ pulmonary venous pressure → pulmonary venous distention and transudation of fluid. Presence of hemosiderin-laden macrophages (“HF” cells) in lungs.

Right heart failure**Hepatomegaly (nutmeg liver)**

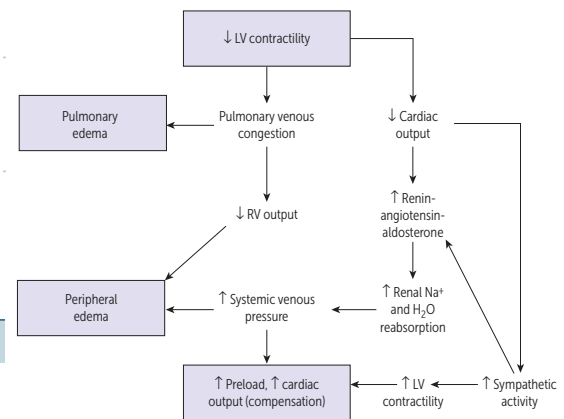
↑ central venous pressure → ↑ resistance to portal flow. Rarely, leads to “cardiac cirrhosis.”

Jugular venous distention

↑ venous pressure.

Peripheral edema

↑ venous pressure → fluid transudation.



Shock

Inadequate organ perfusion and delivery of nutrients necessary for normal tissue and cellular function. Initially may be reversible but life threatening if not treated promptly.

	CAUSED BY	SKIN	PCWP (PRELOAD)	CO	SVR (AFTERLOAD)	TREATMENT
Hypovolemic	Hemorrhage, dehydration, burns	Cold, clammy	↓↓	↓	↑	IV fluids
Cardiogenic	Acute MI, HF, valvular dysfunction, arrhythmia					Inotropes, diuresis
Obstructive	Cardiac tamponade, pulmonary embolism, tension pneumothorax	Cold, clammy	↑ or ↓	↓↓	↑	Relieve obstruction
Distributive	Sepsis, anaphylaxis CNS injury	Warm Dry	↓ ↓	↑ ↓	↓↓ ↓↓	IV fluids, pressors

Bacterial endocarditis

Fever (most common symptom), new murmur, Roth spots (round white spots on retina surrounded by hemorrhage **A**), Osler nodes (tender raised lesions on finger or toe pads **B** due to immune complex deposition), Janeway lesions (small, painless, erythematous lesions on palm or sole) **C**, glomerulonephritis, septic arterial or pulmonary emboli, splinter hemorrhages **D** on nail bed. Multiple blood cultures necessary for diagnosis.

- **Acute**—*S aureus* (high virulence). Large vegetations on previously normal valves **E**. Rapid onset.
- **Subacute**—viridans streptococci (low virulence). Smaller vegetations on congenitally abnormal or diseased valves. Sequela of dental procedures. Gradual onset.

S bovis (*gallolyticus*) is present in colon cancer, *S epidermidis* on prosthetic valves.

Endocarditis may also be nonbacterial (marantic/thrombotic) 2° to malignancy, hypercoagulable state, or lupus.

Mitral valve is most frequently involved.

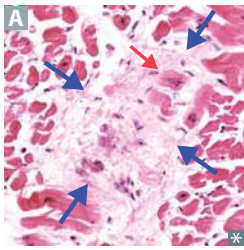
Tricuspid valve endocarditis is associated with IV **drug** abuse (don't "**tri**" **drugs**). Associated with *S aureus*, *Pseudomonas*, and *Candida*.

Culture ⊖; most likely *Coxiella burnetii*, *Bartonella* spp., HACEK (*Haemophilus*, *Aggregatibacter* (formerly *Actinobacillus*), *Cardiobacterium*, *Eikenella*, *Kingella*)

♥ Bacteria **FROM JANE** ♥:

Fever
Roth spots
Osler nodes
Murmur
Janeway lesions
Anemia
Nail-bed hemorrhage
Emboli



Rheumatic fever

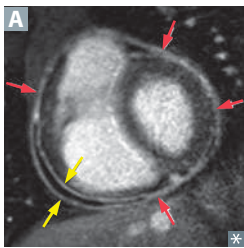
A consequence of pharyngeal infection with group A β -hemolytic streptococci. Late sequelae include rheumatic heart disease, which affects heart valves—mitral > aortic >> tricuspid (high-pressure valves affected most). Early lesion is mitral valve regurgitation; late lesion is mitral stenosis. Associated with Aschoff bodies (granuloma with giant cells [blue arrows in **A**]), Anitschkow cells (enlarged macrophages with ovoid, wavy, rod-like nucleus [red arrow in **A**]), $\uparrow$ anti-streptolysin O (ASO) titers.

Immune mediated (type II hypersensitivity); not a direct effect of bacteria. Antibodies to M protein cross-react with self antigens (molecular mimicry).

Treatment/prophylaxis: penicillin.

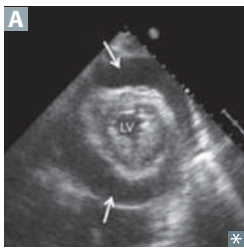
JONES (major criteria):

- Joint (migratory polyarthrititis)
- ♥ (carditis)
- Nodules in skin (subcutaneous)
- Erythema marginatum
- Sydenham chorea

Acute pericarditis

Inflammation of the pericardium [**A**, red arrows]. Commonly presents with sharp pain, aggravated by inspiration, and relieved by sitting up and leaning forward. Often complicated by pericardial effusion [between yellow arrows in **A**]. Presents with friction rub. ECG changes include widespread ST-segment elevation and/or PR depression.

Causes include idiopathic (most common; presumed viral), confirmed infection (eg, Coxsackievirus), neoplasia, autoimmune (eg, SLE, rheumatoid arthritis), uremia, cardiovascular (acute STEMI or Dressler syndrome), radiation therapy.

Cardiac tamponade

Compression of the heart by fluid (eg, blood, effusions [arrows in **A**] in pericardial space) $\rightarrow$ $\downarrow$ CO. Equilibration of diastolic pressures in all 4 chambers.

Findings: Beck triad (hypotension, distended neck veins, distant heart sounds), $\uparrow$ HR, pulsus paradoxus. ECG shows low-voltage QRS and electrical alternans (due to “swinging” movement of heart in large effusion).

Pulsus paradoxus— $\downarrow$ in amplitude of systolic BP by > 10 mm Hg during inspiration. Seen in cardiac tamponade, asthma, obstructive sleep apnea, pericarditis, croup.

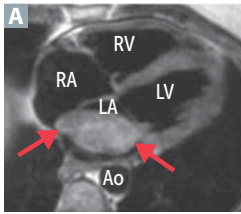
Syphilitic heart disease

3° syphilis disrupts the vasa vasorum of the aorta with consequent atrophy of vessel wall and dilatation of aorta and valve ring. May see calcification of aortic root, ascending aortic arch, and thoracic aorta. Leads to “tree bark” appearance of aorta.

Can result in aneurysm of ascending aorta or aortic arch, aortic insufficiency.

Cardiac tumors

Most common heart tumor is a metastasis (eg, melanoma).

Myxomas

Most common 1° cardiac tumor (red arrows) in adults **A**. 90% occur in the atria (mostly left atrium). Myxomas are usually described as a “ball valve” obstruction in the left atrium (associated with multiple syncopal episodes). May auscultate early diastolic “tumor plop” sound. Histology: gelatinous material, myxoma cells immersed in glycosaminoglycans.

Rhabdomyomas

Most frequent 1° cardiac tumor in children (associated with tuberous sclerosis). Histology: hamartomatous growths.

Kussmaul sign

↑ in JVP on inspiration instead of a normal ↓.

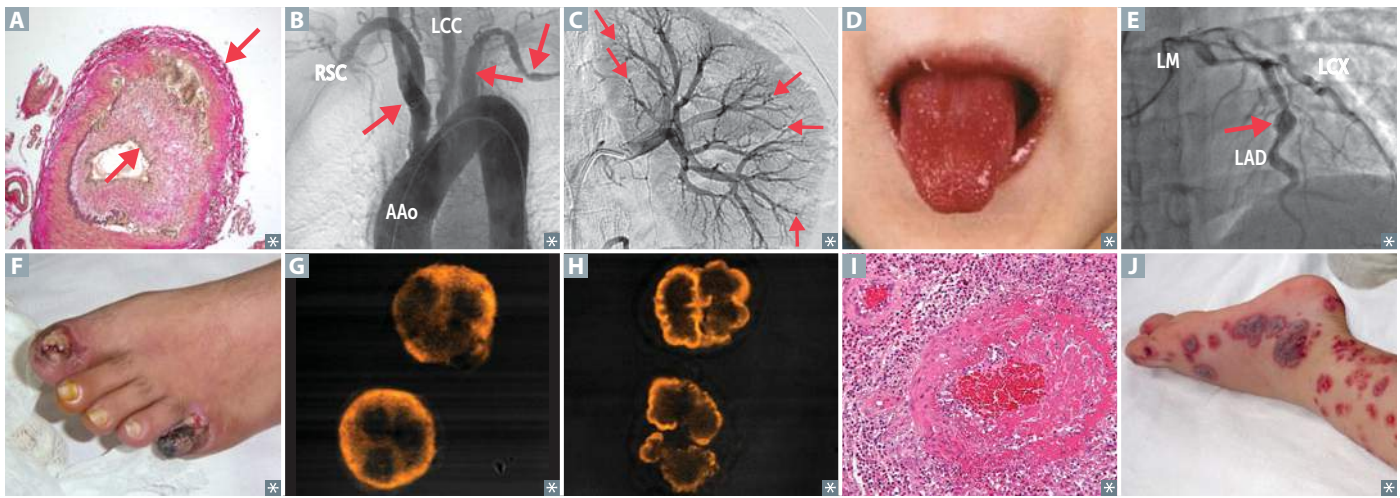
Inspiration → negative intrathoracic pressure not transmitted to heart → impaired filling of right ventricle → blood backs up into venae cavae → JVD. May be seen with constrictive pericarditis, restrictive cardiomyopathies, right atrial or ventricular tumors.

Vasculitides

	EPIDEMIOLOGY/PRESENTATION	PATHOLOGY/LABS
Large-vessel vasculitis		
Giant cell (temporal) arteritis	Usually elderly females. Unilateral headache (temporal artery), jaw claudication. May lead to irreversible blindness due to ophthalmic artery occlusion. Associated with polymyalgia rheumatica.	Most commonly affects branches of carotid artery. Focal granulomatous inflammation A . ↑ ESR. Treat with high-dose corticosteroids prior to temporal artery biopsy to prevent blindness.
Takayasu arteritis	Usually Asian females < 40 years old. “Pulseless disease” (weak upper extremity pulses), fever, night sweats, arthritis, myalgias, skin nodules, ocular disturbances.	Granulomatous thickening and narrowing of aortic arch and proximal great vessels B . ↑ ESR. Treat with corticosteroids.
Medium-vessel vasculitis		
Polyarteritis nodosa	Usually middle-aged men. Hepatitis B seropositivity in 30% of patients. Fever, weight loss, malaise, headache. GI: abdominal pain, melena. Hypertension, neurologic dysfunction, cutaneous eruptions, renal damage.	Typically involves renal and visceral vessels, not pulmonary arteries. Transmural inflammation of the arterial wall with fibrinoid necrosis. Different stages of inflammation may coexist in different vessels. Innumerable renal microaneurysms C and spasms on arteriogram. Treat with corticosteroids, cyclophosphamide.
Kawasaki disease (mucocutaneous lymph node syndrome)	Asian children < 4 years old. C onjunctival injection, R ash (polymorphous → desquamating), A denopathy (cervical), S trawberry tongue (oral mucositis) D , H and-foot changes (edema, erythema), f ever.	CRASH and burn . May develop coronary artery aneurysms E ; thrombosis or rupture can cause death. Treat with IV immunoglobulin and aspirin.
Buerger disease (thromboangiitis obliterans)	Heavy smokers, males < 40 years old. Intermittent claudication may lead to gangrene F , autoamputation of digits, superficial nodular phlebitis. Raynaud phenomenon is often present.	Segmental thrombosing vasculitis. Treat with smoking cessation.
Small-vessel vasculitis		
Granulomatosis with polyangiitis (Wegener)	Upper respiratory tract: perforation of nasal septum, chronic sinusitis, otitis media, mastoiditis. Lower respiratory tract: hemoptysis, cough, dyspnea. Renal: hematuria, red cell casts.	Triad: ▪ Focal necrotizing vasculitis ▪ Necrotizing granulomas in the lung and upper airway ▪ Necrotizing glomerulonephritis PR3-ANCA/c-ANCA G (anti-proteinase 3). CXR: large nodular densities. Treat with cyclophosphamide, corticosteroids.
Microscopic polyangiitis	Necrotizing vasculitis commonly involving lung, kidneys, and skin with pauci-immune glomerulonephritis and palpable purpura. Presentation similar to granulomatosis with polyangiitis but without nasopharyngeal involvement.	No granulomas. MPO-ANCA/p-ANCA H (anti-myeloperoxidase). Treat with cyclophosphamide, corticosteroids.

Vasculitides (continued)

	EPIDEMIOLOGY/PRESENTATION	PATHOLOGY/LABS
Small-vessel vasculitis (continued)		
Eosinophilic granulomatosis with polyangiitis (Churg-Strauss)	Asthma, sinusitis, skin nodules or purpura, peripheral neuropathy (eg, wrist/foot drop). Can also involve heart, GI, kidneys (pauci-immune glomerulonephritis).	Granulomatous, necrotizing vasculitis with eosinophilia I . MPO-ANCA/p-ANCA, ↑ IgE level.
Henoch-Schönlein purpura	Most common childhood systemic vasculitis. Often follows URI. Classic triad: ▪ Skin: palpable purpura on buttocks/legs J ▪ Arthralgias ▪ GI: abdominal pain	Vasculitis 2° to IgA immune complex deposition. Associated with IgA nephropathy (Berger disease).

**Hereditary hemorrhagic telangiectasia**

Inherited disorder of blood vessels. Findings: blanching skin lesions (telangiectasias), recurrent epistaxis, skin discolorations, arteriovenous malformations (AVMs), GI bleeding, hematuria. Also known as Osler-Weber-Rendu syndrome.

► CARDIOVASCULAR—PHARMACOLOGY

Hypertension treatment

Primary (essential) hypertension	Thiazide diuretics, ACE inhibitors, angiotensin II receptor blockers (ARBs), dihydropyridine Ca^{2+} channel blockers.	
Hypertension with heart failure	Diuretics, ACE inhibitors/ARBs, β -blockers (compensated HF), aldosterone antagonists.	β -blockers must be used cautiously in decompensated HF and are contraindicated in cardiogenic shock.
Hypertension with diabetes mellitus	ACE inhibitors/ARBs, Ca^{2+} channel blockers, thiazide diuretics, β -blockers.	ACE inhibitors/ARBs are protective against diabetic nephropathy.
Hypertension in pregnancy	Hydralazine, labetalol, methyldopa, nifedipine.	

Calcium channel blockers

MECHANISM	Block voltage-dependent L-type calcium channels of cardiac and smooth muscle $\rightarrow$ $\downarrow$ muscle contractility. Vascular smooth muscle—amlodipine = nifedipine > diltiazem > verapamil. Heart—verapamil > diltiazem > amlodipine = nifedipine (verapamil = ventricle).	
CLINICAL USE	Dihydropyridines (except nimodipine): hypertension, angina (including Prinzmetal), Raynaud phenomenon. Nimodipine: subarachnoid hemorrhage (prevents cerebral vasospasm). Nicardipine, clevidipine: hypertensive urgency or emergency. Non-dihydropyridines: hypertension, angina, atrial fibrillation/flutter.	
ADVERSE EFFECTS	Non-dihydropyridine: cardiac depression, AV block, hyperprolactinemia, constipation. Dihydropyridine: peripheral edema, flushing, dizziness, gingival hyperplasia.	

Hydralazine

MECHANISM	$\uparrow$ cGMP $\rightarrow$ smooth muscle relaxation. Vasodilates arterioles > veins; afterload reduction.
CLINICAL USE	Severe hypertension (particularly acute), HF (with organic nitrate). Safe to use during pregnancy. Frequently coadministered with a β -blocker to prevent reflex tachycardia.
ADVERSE EFFECTS	Compensatory tachycardia (contraindicated in angina/CAD), fluid retention, headache, angina. Lupus-like syndrome.

Hypertensive emergency

Nitroprusside	Short acting; $\uparrow$ cGMP via direct release of NO. Can cause cyanide toxicity (releases cyanide).
Fenoldopam	Dopamine D_1 receptor agonist—coronary, peripheral, renal, and splanchnic vasodilation. $\downarrow$ BP, $\uparrow$ natriuresis. Also used postoperatively as an antihypertensive. Can cause hypotension and tachycardia.

Nitrates	Nitroglycerin, isosorbide dinitrate, isosorbide mononitrate.
MECHANISM	Vasodilate by ↑ NO in vascular smooth muscle → ↑ in cGMP and smooth muscle relaxation. Dilate veins >> arteries. ↓ preload.
CLINICAL USE	Angina, acute coronary syndrome, pulmonary edema.
ADVERSE EFFECTS	Reflex tachycardia (treat with β-blockers), hypotension, flushing, headache, “Monday disease” in industrial exposure: development of tolerance for the vasodilating action during the work week and loss of tolerance over the weekend → tachycardia, dizziness, headache upon reexposure. Contraindicated in right ventricular infarction.

Antianginal therapy Goal is reduction of myocardial O₂ consumption (MVO₂) by ↓ 1 or more of the determinants of MVO₂: end-diastolic volume, BP, HR, contractility.

COMPONENT	NITRATES	β-BLOCKERS	NITRATES + β-BLOCKERS
End-diastolic volume	↓	No effect or ↑	No effect or ↓
Blood pressure	↓	↓	↓
Contractility	No effect	↓	Little/no effect
Heart rate	↑ (reflex response)	↓	No effect or ↓
Ejection time	↓	↑	Little/no effect
MVO ₂	↓	↓	↓↓

Verapamil is similar to β-blockers in effect.

Pindolol and acebutolol are partial β-agonists that should be used with caution in angina.

Ranolazine

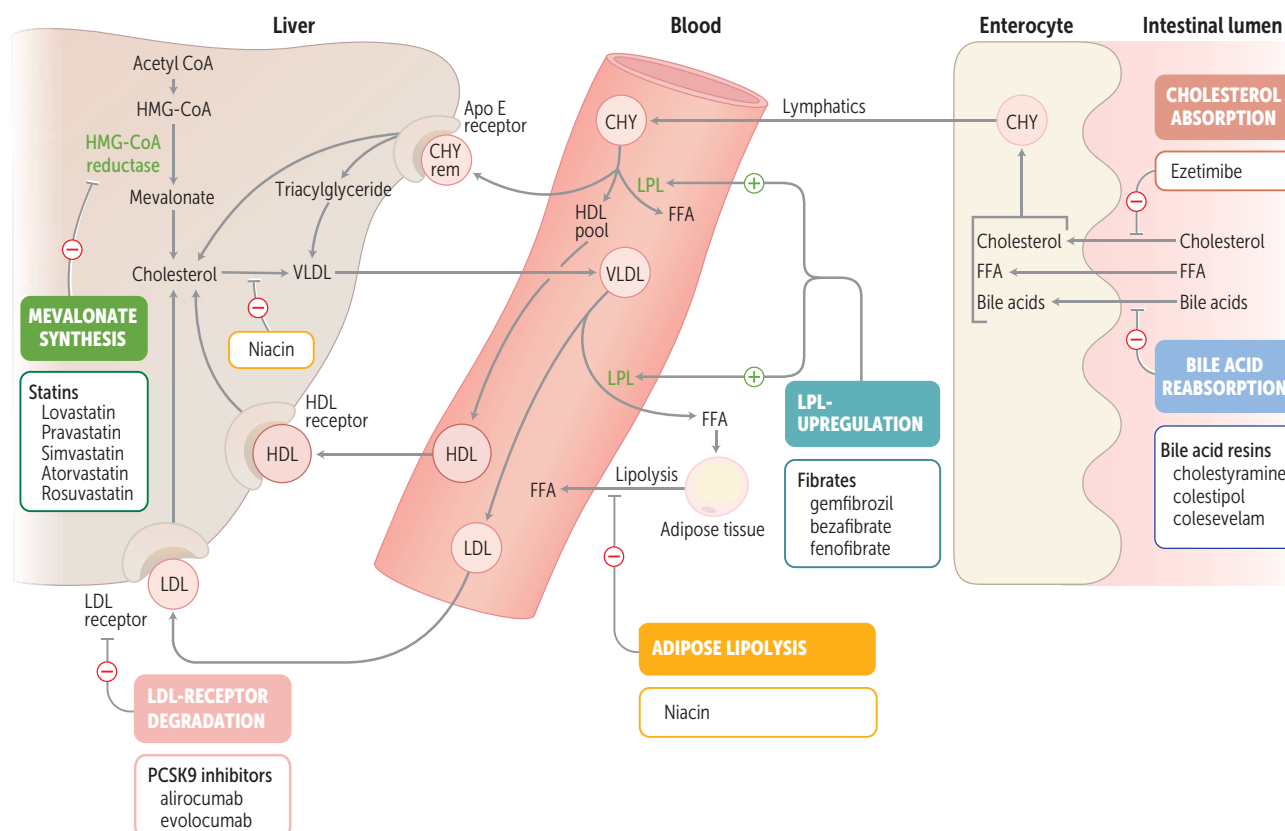
MECHANISM	Inhibits the late phase of sodium current thereby reducing diastolic wall tension and oxygen consumption. Does not affect heart rate or contractility.
CLINICAL USE	Angina refractory to other medical therapies.
ADVERSE EFFECTS	Constipation, dizziness, headache, nausea, QT prolongation.

Milrinone

MECHANISM	Selective PDE-3 inhibitor. In cardiomyocytes: ↑ cAMP accumulation → ↑ Ca ²⁺ influx → ↑ inotropy and chronotropy. In vascular smooth muscle: ↑ cAMP accumulation → inhibition of MLCK activity → general vasodilation.
CLINICAL USE	Short-term use in acute decompensated HF.
ADVERSE EFFECTS	Arrhythmias, hypotension.

Lipid-lowering agents

DRUG	LDL	HDL	TRIGLYCERIDES	MECHANISMS OF ACTION	ADVERSE EFFECTS/PROBLEMS
HMG-CoA reductase inhibitors (eg, lovastatin, pravastatin)	↓↓↓	↑	↓	Inhibit conversion of HMG-CoA to mevalonate, a cholesterol precursor; ↓ mortality in CAD patients	Hepatotoxicity (↑ LFTs), myopathy (esp. when used with fibrates or niacin)
Bile acid resins Cholestyramine, colestipol, colesevelam	↓↓	Slightly ↑	Slightly ↑	Prevent intestinal reabsorption of bile acids; liver must use cholesterol to make more	GI upset, ↓ absorption of other drugs and fat-soluble vitamins
Ezetimibe	↓↓	↑/—	↓/—	Prevent cholesterol absorption at small intestine brush border	Rare ↑ LFTs, diarrhea
Fibrates Gemfibrozil, bezafibrate, fenofibrate	↓	↑	↓↓↓	Upregulate LPL → ↑ TG clearance Activates PPAR- α to induce HDL synthesis	Myopathy (↑ risk with statins), cholesterol gallstones
Niacin (vitamin B₃)	↓↓	↑↑	↓	Inhibits lipolysis (hormone- sensitive lipase) in adipose tissue; reduces hepatic VLDL synthesis	Red, flushed face, which is ↓ by NSAIDs or long-term use Hyperglycemia Hyperuricemia
PCSK9 inhibitors Alirocumab, evolocumab	↓↓↓	↑	↓	Inactivation of LDL-receptor degradation, increasing amount of LDL removed from bloodstream	Myalgias, delirium, dementia, other neurocognitive effects

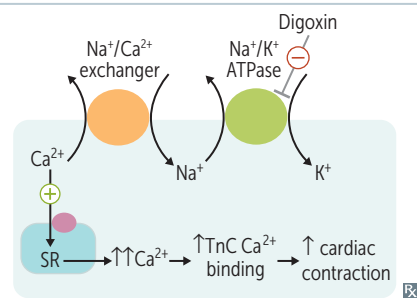


Cardiac glycosides

Digoxin.

MECHANISM

Direct inhibition of Na^+/K^+ ATPase
 → indirect inhibition of $\text{Na}^+/\text{Ca}^{2+}$ exchanger.
 $\uparrow [\text{Ca}^{2+}]_i \rightarrow$ positive inotropy. Stimulates vagus nerve $\rightarrow \downarrow$ HR.

**CLINICAL USE**

HF ($\uparrow$ contractility); atrial fibrillation ($\downarrow$ conduction at AV node and depression of SA node).

ADVERSE EFFECTS

Cholinergic—nausea, vomiting, diarrhea, blurry yellow vision (think van Gogh), arrhythmias, AV block.

Can lead to hyperkalemia, which indicates poor prognosis.

Factors predisposing to toxicity: renal failure ($\downarrow$ excretion), hypokalemia (permissive for digoxin binding at K^+ -binding site on Na^+/K^+ ATPase), drugs that displace digoxin from tissue-binding sites, and $\downarrow$ clearance (eg, verapamil, amiodarone, quinidine).

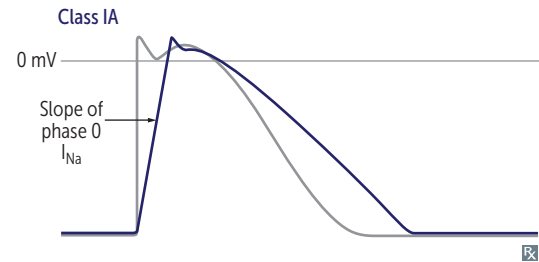
ANTIDOTE

Slowly normalize K^+ , cardiac pacer, anti-digoxin Fab fragments, Mg^{2+} .

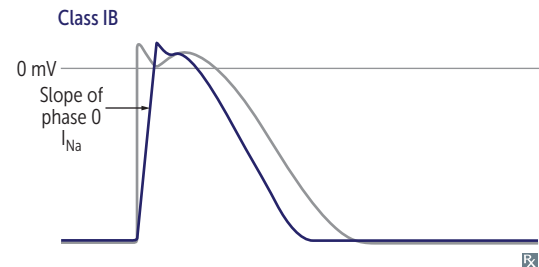
Antiarrhythmics— sodium channel blockers (class I)

Slow or block (↓) conduction (especially in depolarized cells). ↓ slope of phase 0 depolarization. Are state dependent (selectively depress tissue that is frequently depolarized [eg, tachycardia]).

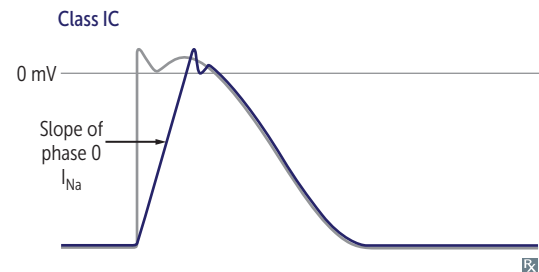
Class IA	Quinidine, Procainamide, Disopyramide. “The Q ueen P roclaims D iso’s p yramid.”
MECHANISM	↑ AP duration, ↑ effective refractory period (ERP) in ventricular action potential, ↑ QT interval, some potassium channel blocking effects.
CLINICAL USE	Both atrial and ventricular arrhythmias, especially re-entrant and ectopic SVT and VT.
ADVERSE EFFECTS	Cinchonism (headache, tinnitus with quinidine), reversible SLE-like syndrome (procainamide), HF (disopyramide), thrombocytopenia, torsades de pointes due to ↑ QT interval.



Class IB	Lidocaine, Mexiletine. “I’d B uy L iddy’s M exican T acos.”
MECHANISM	↓ AP duration. Preferentially affect ischemic or depolarized Purkinje and ventricular tissue. Phenytoin can also fall into the IB category.
CLINICAL USE	Acute ventricular arrhythmias (especially post-MI), digitalis-induced arrhythmias. IB is B est post-MI.
ADVERSE EFFECTS	CNS stimulation/depression, cardiovascular depression.



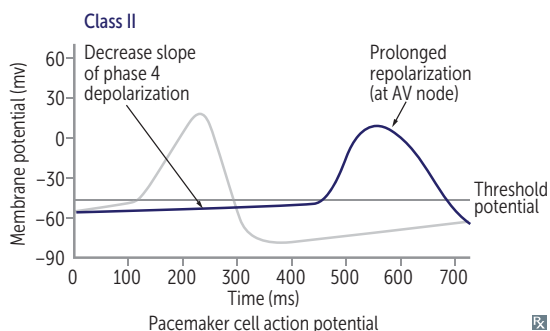
Class IC	Flecainide, Propafenone. “Can I have F ries, P lease.”
MECHANISM	Significantly prolongs ERP in AV node and accessory bypass tracts. No effect on ERP in Purkinje and ventricular tissue. Minimal effect on AP duration.
CLINICAL USE	SVTs, including atrial fibrillation. Only as a last resort in refractory VT.
ADVERSE EFFECTS	Proarrhythmic, especially post-MI (contraindicated). IC is C ontraindicated in structural and ischemic heart disease.



**Antiarrhythmics—
β-blockers (class II)**

Metoprolol, propranolol, esmolol, atenolol, timolol, carvedilol.

MECHANISM	Decrease SA and AV nodal activity by ↓ cAMP, ↓ Ca^{2+} currents. Suppress abnormal pacemakers by ↓ slope of phase 4. AV node particularly sensitive—↑ PR interval. Esmolol very short acting.
CLINICAL USE	SVT, ventricular rate control for atrial fibrillation and atrial flutter.
ADVERSE EFFECTS	Impotence, exacerbation of COPD and asthma, cardiovascular effects (bradycardia, AV block, HF), CNS effects (sedation, sleep alterations). May mask the signs of hypoglycemia. Metoprolol can cause dyslipidemia. Propranolol can exacerbate vasospasm in Prinzmetal angina. β-blockers (except the nonselective α- and β-antagonists carvedilol and labetalol) cause unopposed α ₁ -agonism if given alone for pheochromocytoma or cocaine toxicity. Treat β-blocker overdose with saline, atropine, glucagon.

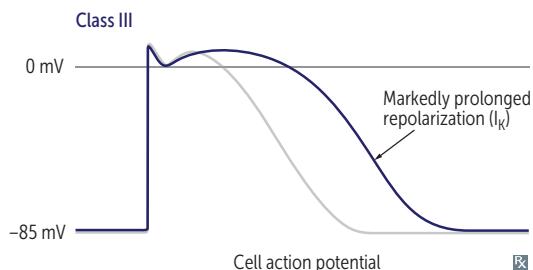
**Antiarrhythmics—
potassium channel
blockers (class III)**

Amiodarone, Ibutilide, Dofetilide, Sotalol.

AIDS.

MECHANISM	↑ AP duration, ↑ ERP, ↑ QT interval.
CLINICAL USE	Atrial fibrillation, atrial flutter; ventricular tachycardia (amiodarone, sotalol).
ADVERSE EFFECTS	Sotalol—torsades de pointes, excessive β blockade. Ibutilide—torsades de pointes. Amiodarone—pulmonary fibrosis, hepatotoxicity, hypothyroidism/hyperthyroidism (amiodarone is 40% iodine by weight), acts as hapten (corneal deposits, blue/gray skin deposits resulting in photodermatitis), neurologic effects, constipation, cardiovascular effects (bradycardia, heart block, HF).

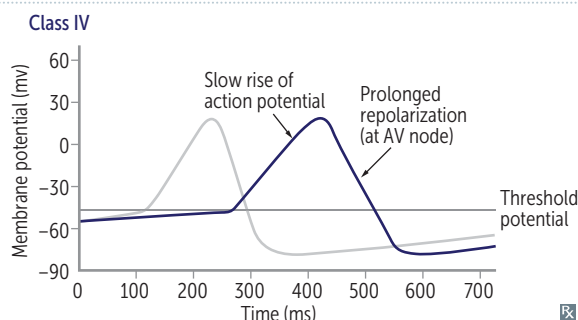
Remember to check PFTs, LFTs, and TFTs when using amiodarone.
Amiodarone is lipophilic and has class I, II, III, and IV effects.



Antiarrhythmics— calcium channel blockers (class IV)

Verapamil, diltiazem.

MECHANISM	↓ conduction velocity, ↑ ERP, ↑ PR interval.
CLINICAL USE	Prevention of nodal arrhythmias (eg, SVT), rate control in atrial fibrillation.
ADVERSE EFFECTS	Constipation, flushing, edema, cardiovascular effects (HF, AV block, sinus node depression).



Other antiarrhythmics

Adenosine ↑ K^+ out of cells → hyperpolarizing the cell and ↓ I_{Ca} , decreasing AV node conduction. Drug of choice in diagnosing/terminating certain forms of SVT. Very short acting (~ 15 sec). Effects blunted by theophylline and caffeine (both are adenosine receptor antagonists). Adverse effects include flushing, hypotension, chest pain, sense of impending doom, bronchospasm.

Mg^{2+} Effective in torsades de pointes and digoxin toxicity.

Ivabradine

MECHANISM	Selective inhibition of funny sodium channels (I_f), prolonging slow depolarization phase (phase 4). ↓ SA node firing; negative chronotropic effect without inotropy. Reduces cardiac O_2 requirement.
CLINICAL USE	Chronic stable angina in patients who cannot take β -blockers. Chronic HF with reduced ejection fraction.
ADVERSE EFFECTS	Luminous phenomena/visual brightness, hypertension, bradycardia.